

Multivariate Analysis of Milk Transportation Cost

Contents

1	Introduction	3
1.1	Background and Context	3
1.2	Description of the Dataset	3
1.3	Objectives of the Study	4
1.4	Analytical Issues to be Examined Later	4
1.5	Organization of the Report	5
2	Data Description and Exploratory Analysis	6
2.1	Overview of the Data	6
2.2	Basic Numerical Summaries	6
2.3	Univariate Visual Exploration	7
2.4	Multivariate Visual Exploration	9
2.5	Correlation and Dependence Structure	11
2.6	Initial Structural Insights	12
2.7	Early Evidence of Outliers and Skewness	12
3	Assumption Checking and Data Diagnostics	13
3.1	Purpose of the Diagnostic Stage	13
3.2	Detection of Suspicious Observations	13
3.3	Normality Assessment	16
3.3.1	Univariate normality	16
3.3.2	Multivariate normality	16
3.3.3	Interpretation	17
3.4	Effect of Removing Suspicious Observations	17
3.4.1	Candidate deletion rules	17
3.4.2	Results after removing rows 9 and 21	18
3.4.3	Results after removing rows 9, 21, 41, 47, and 56	19
3.4.4	Comparison of the two deletion strategies	20
3.4.5	Implication for later analysis	20
3.5	Effect of Transformations	20
3.5.1	Aim of the transformation analysis	20
3.5.2	Transformations considered	21
3.5.3	Effect on univariate normality	21
3.5.4	Effect on multivariate normality	23
3.5.5	Comparison across transformations	24

3.5.6	Recovering overall normality	24
3.6	Regression-Based Imputation of Suspicious Observations	25
3.6.1	Univariate normality after imputation	25
3.6.2	Multivariate normality after imputation	27
3.6.3	Conclusion	28
3.7	Final Preprocessing Decision	28
4	Two-Sample Multivariate Inference for Gasoline and Diesel Fleets	30
4.1	Inferential Setup	30
4.2	Descriptive Multivariate Summaries	30
4.3	Equality of Covariance Matrices	31
4.4	Equality of Mean Vectors	32
4.5	Simultaneous Confidence Intervals for Mean Differences	32
4.6	Graphical Synthesis of Group Differences	33
5	LDA and QDA	36
5.1	Objective and datasets considered	36
5.2	LDA: cross-validated classification performance	37
5.3	LDA decision-boundary plots	37
5.4	QDA: cross-validated classification performance	38
5.5	QDA decision-boundary plots	39
6	PCA and Factor Analysis	41
6.1	Suitability for Principal Component and Factor Analysis	41
6.2	Principal Component Analysis	41
6.3	Exploratory Factor Analysis	43
A	Normality Tests Used	45
A.1	Univariate Normality Tests	45
A.1.1	Shapiro–Wilk Test	45
A.1.2	Anderson–Darling Test	45
A.1.3	Lilliefors Test	46
A.2	Multivariate Normality Tests	46
A.2.1	Mardia’s Skewness and Kurtosis	46
A.2.2	Henze–Zirkler Test	47
A.2.3	Royston Test	47
A.2.4	Doornik–Hansen Test	47
A.3	Interpretation	48

Chapter 1

Introduction

1.1 Background and Context

The dataset considered in this report arises from a study of the cost of transporting milk from farms to dairy plants. For each transport unit, three cost components are recorded on a per-mile basis: fuel cost, repair cost, and capital cost. In addition, each observation is classified according to fuel type, namely gasoline or diesel.

This setting is naturally multivariate. The economic structure of transportation cost is not captured by any single variable alone; rather, it depends on the joint behaviour of several related cost components. A separate univariate treatment of fuel, repair, and capital cost would ignore possible association among them and may fail to describe the overall cost pattern adequately.

The presence of two fuel types also makes the data suitable for comparative study. One may ask whether gasoline- and diesel-based transport systems differ in their overall cost profiles, whether their variability and dependence structures are similar, and whether the observed cost variables can be used to distinguish between the two groups.

Thus, the dataset provides a useful framework for applying core tools of multivariate analysis. The report studies the joint cost structure, explores differences between the two fuel types, and uses the data as a basis for inference, comparison, and classification.

1.2 Description of the Dataset

The data comes from a study on the cost of transporting milk from farms to dairy plants.

Variable	Description	Unit
FuelType	Type of fuel used by the vehicle	Categorical
X_1	Fuel cost	cents/mi
X_2	Repair cost	cents/mi
X_3	Capital cost	cents/mi

Table 1.1: Variables recorded in the milk transportation dataset.

For each observation, transportation cost is recorded through three per-mile components, along with the fuel type of the vehicle. The dataset is stored in `milk_transportation.txt`. In the raw file, `FuelType` is coded as 1 = gasoline and 2 = diesel; throughout the report, these are treated as the two observed groups under study.

The dataset contains 59 observations in total: 36 in the gasoline group and 23 in the diesel group. Thus, each observation may be viewed as a three-dimensional cost vector together with a group label.

1.3 Objectives of the Study

The main objectives of this report are:

1. to study the joint behaviour of the three transportation cost variables X_1 , X_2 , and X_3 ;
2. to compare the gasoline and diesel groups in terms of their multivariate cost structure;
3. to carry out formal multivariate inference on means, covariance structure, and group differences;
4. to examine association among the variables through correlation- and regression-based methods;
5. to study the problem of classifying observations by fuel type using discriminant methods; and
6. to explore lower-dimensional structure in the data through principal component analysis and factor analysis.

These objectives are pursued using the milk transportation dataset, in which each observation consists of three cost variables together with a fuel-type label.

1.4 Analytical Issues to be Examined Later

Before applying formal multivariate methods, several analytical issues must be examined carefully.

First, the adequacy of distributional assumptions must be assessed. Many classical procedures in multivariate analysis are developed under multivariate normality, so it is necessary to examine normality both for the data as a whole and within the observed fuel-type groups.

Second, the data must be checked for unusual or influential observations. Since the dataset is low-dimensional and group comparisons play a central role, even a small number of extreme points may materially affect estimates, test statistics, confidence regions, and classification rules.

Third, the possibility of unequal covariance across groups must be kept in view. Some methods require or assume a common covariance matrix, whereas others allow group-specific covariance patterns. This issue therefore has direct implications for model choice.

Fourth, preprocessing decisions must be considered. These include whether transformations are needed, whether certain observations should be treated separately, and whether the same version of the dataset should be used for all analyses.

1.5 Organization of the Report

The remainder of the report is organized as follows. The next chapter presents an exploratory analysis of the dataset and summarizes its main empirical features. This is followed by a chapter on assumption checking and preprocessing, where issues such as unusual observations, normality, and possible transformations are examined.

Subsequent chapters develop the main inferential and comparative analyses, including procedures for mean vectors, covariance structure, and multivariate comparison of the gasoline and diesel groups. A separate chapter is devoted to discriminant analysis and classification. The final analytical chapter considers dimension-reduction methods, namely principal component analysis and factor analysis.

The report concludes with a summary of the main findings and a brief discussion of the overall multivariate structure of the data.

Chapter 2

Data Description and Exploratory Analysis

2.1 Overview of the Data

The exploratory analysis in this chapter is based on the raw dataset `milk_transportation.txt`, which records transportation costs for milk hauled from farms to dairy plants. Each observation consists of three cost variables measured in cents per mile together with the fuel type of the vehicle: gasoline or diesel.

Feature	Description	Value
Number of observations	Total sample size	59
Number of variables	Fuel type + 3 cost variables	4
Groups	Fuel type categories	Gasoline, Diesel
Group sizes	Gasoline / Diesel	36 / 23
Cost variables	Per-mile components	X_1, X_2, X_3

Table 2.1: Basic structure of the milk transportation dataset.

Thus, the data may be viewed as a collection of three-dimensional cost observations, partitioned according to fuel type. The purpose of the present chapter is to summarize these observations numerically and visually before turning to formal assumption-checking in the next chapter.

2.2 Basic Numerical Summaries

Table 2.2 gives the overall summary of the three cost variables, while Tables 2.3 and 2.4 give the corresponding summaries separately for the gasoline and diesel groups.

Variable	Min	Q1	Median	Mean	Q3	Max	Range	SD
X_1	4.24	9.12	10.28	11.39	12.70	29.11	24.87	4.08
X_2	1.35	5.15	8.89	9.15	12.58	21.52	20.17	4.70
X_3	3.28	8.15	11.23	12.93	17.00	35.18	31.90	6.63

Table 2.2: Overall numerical summaries for the full dataset.

Variable	Min	Q1	Median	Mean	Q3	Max	Range	SD
X_1	4.24	9.41	11.16	12.22	13.55	29.11	24.87	4.80
X_2	1.35	5.07	7.67	8.11	11.27	17.44	16.09	4.19
X_3	3.28	6.80	9.62	9.59	11.96	17.59	14.31	3.74

Table 2.3: Numerical summaries for the gasoline group.

Variable	Min	Q1	Median	Mean	Q3	Max	Range	SD
X_1	6.47	8.81	9.77	10.11	11.48	15.90	9.43	2.09
X_2	2.94	5.96	11.75	10.76	13.24	21.52	18.58	5.08
X_3	5.99	14.11	17.45	18.17	21.02	35.18	29.19	6.83

Table 2.4: Numerical summaries for the diesel group.

Several broad features are already visible from these tables. The gasoline group has a higher average fuel cost (X_1), whereas the diesel group has higher repair and capital costs (X_2 and especially X_3). The variability pattern also differs across groups: gasoline shows greater spread in X_1 , while diesel shows greater spread in X_2 and X_3 . In particular, the large range and standard deviation of X_3 in the diesel group suggest substantial heterogeneity in capital cost. Overall, the summaries indicate that the two groups differ in level and dispersion.

2.3 Univariate Visual Exploration

To complement the numerical summaries, Figures 2.1, 2.2, and 2.3 display the marginal behaviour of the three cost variables.

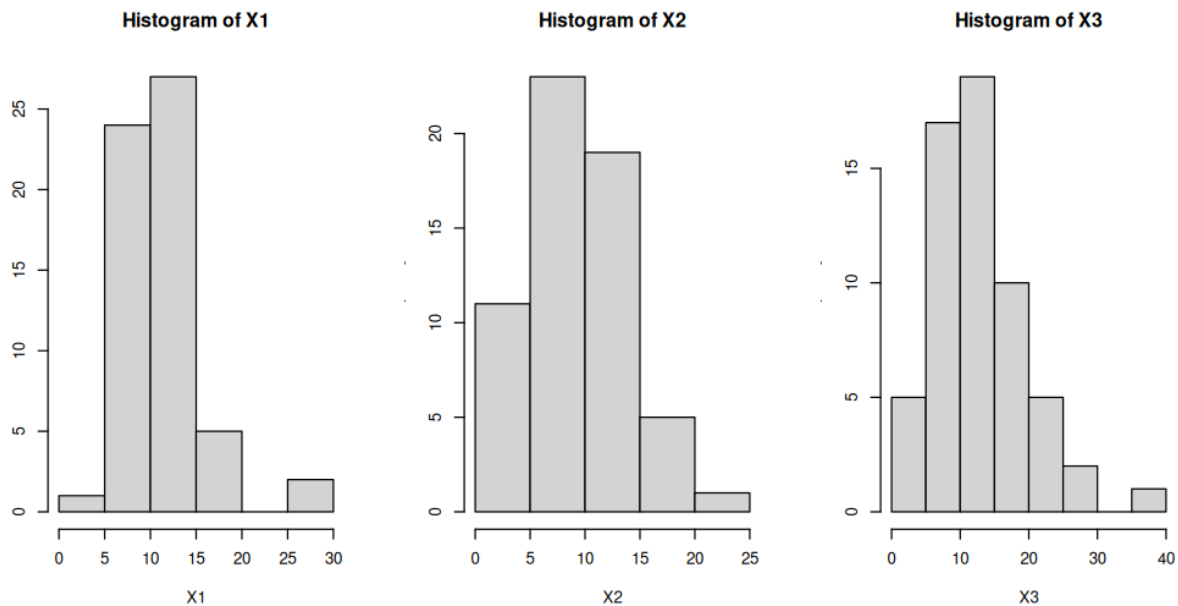


Figure 2.1: Histograms of the three cost variables for the full dataset.

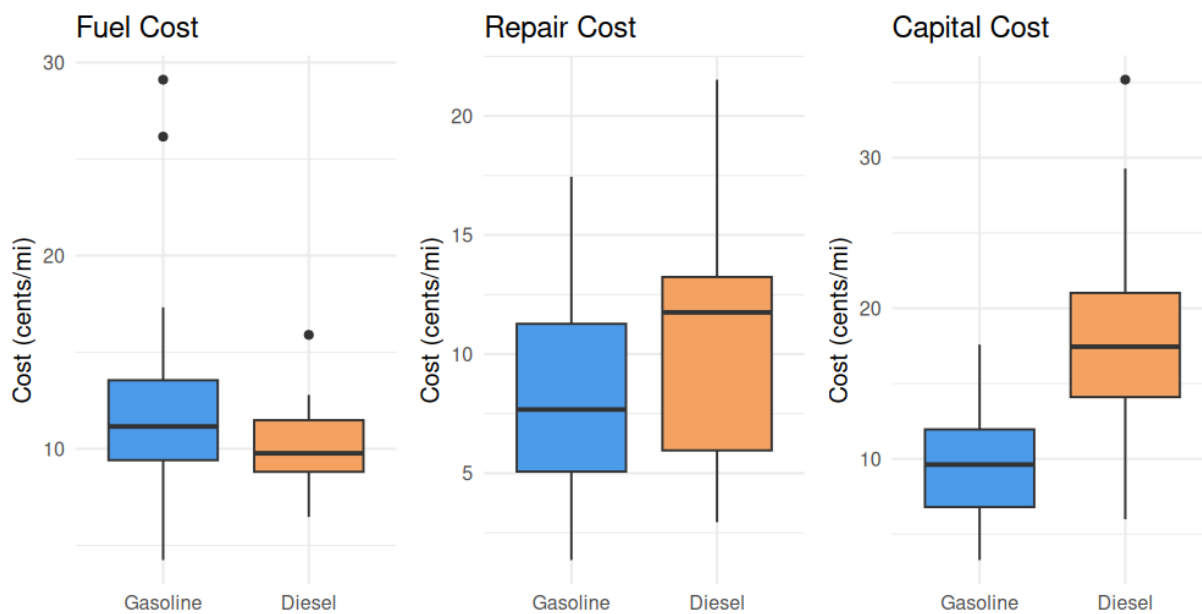


Figure 2.2: Side-by-side boxplots of fuel, repair, and capital cost by fuel type.

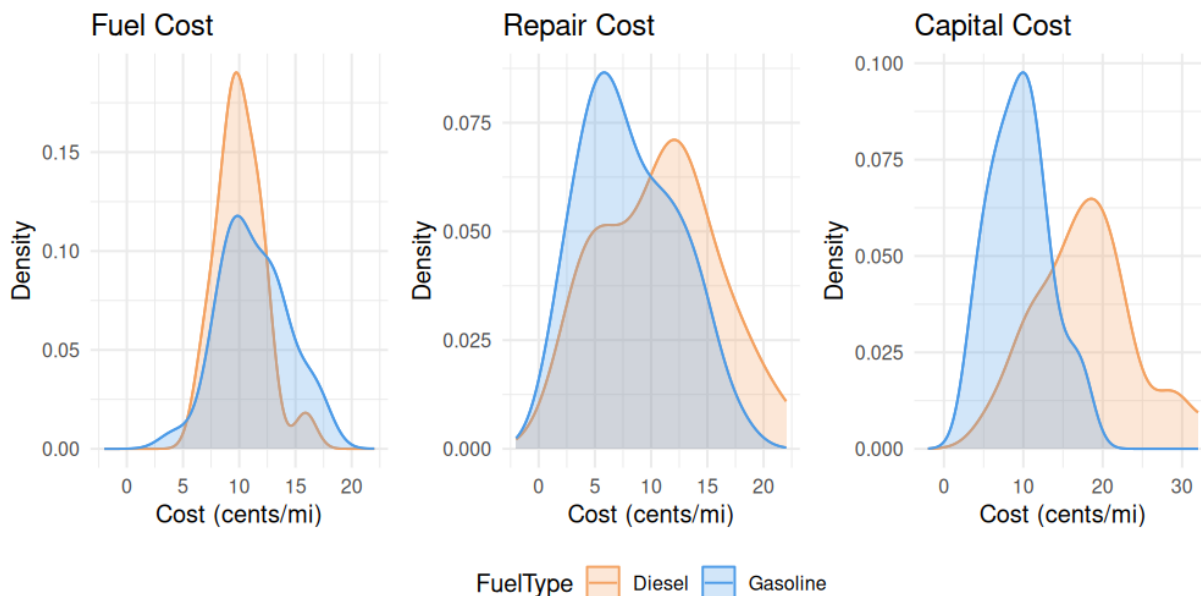


Figure 2.3: Groupwise density plots for the three transportation cost variables.

The histograms provide an initial view of the pooled marginal distributions. Among the three variables, X_1 and X_3 show noticeable right-tailed behaviour, while X_2 appears comparatively less asymmetric. In particular, the long upper tail in X_1 suggests the presence of a few unusually large fuel-cost observations.

The boxplots allow a groupwise comparison of center, spread, and extreme values. For X_1 , the gasoline group appears slightly higher in center and exhibits a longer upper tail, with a few unusually large observations. For X_2 , the diesel group is shifted upward relative to gasoline, though the two groups still overlap substantially. For X_3 , the difference is clearer: diesel observations tend to lie at distinctly higher values than gasoline observations.

The density plots further clarify the extent of overlap between the two groups. The overlap is considerable for X_1 and X_2 , but is smaller for X_3 , where the two densities are more clearly separated. These plots also reinforce the impression of skewness in some variables, especially in the upper tails.

Overall, the univariate plots suggest that the two fuel types differ in marginal location and spread, with the strongest visual separation appearing in capital cost. They also provide early visual evidence of skewness and possible extreme observations, which will be examined more carefully in later sections.

2.4 Multivariate Visual Exploration

To examine the joint structure of the data, Figure 2.4 shows the pairwise scatterplot matrix, while Figure 2.5 displays the observations in three-dimensional cost space.

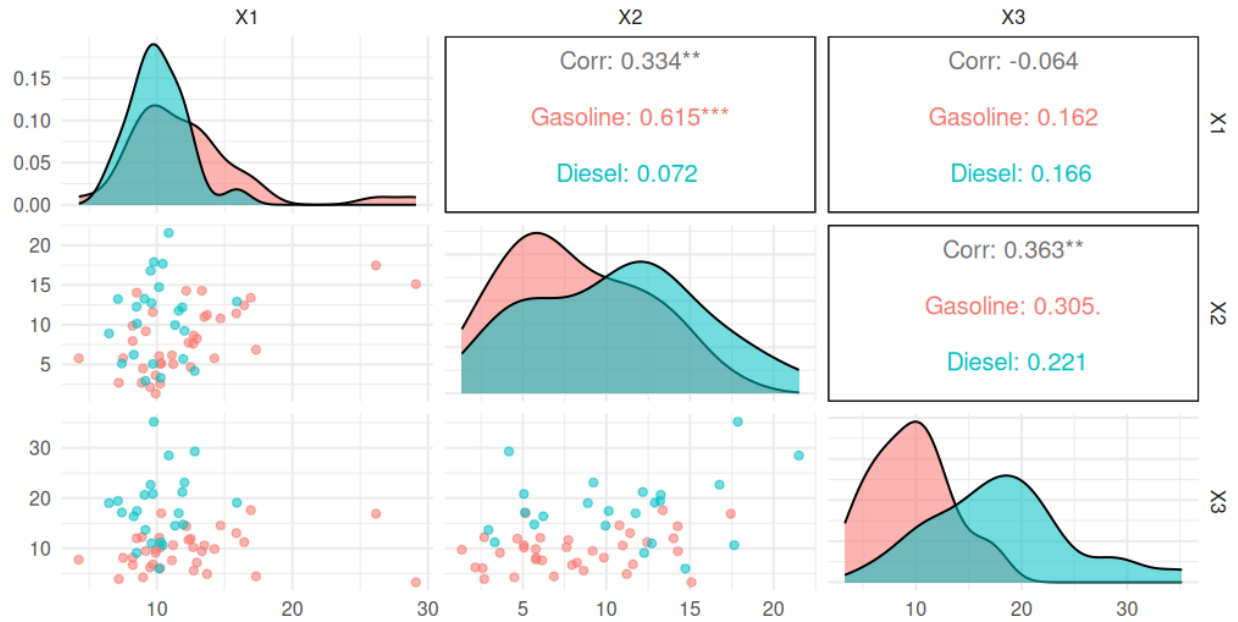


Figure 2.4: Pairwise scatterplot matrix of the three cost variables by fuel type.

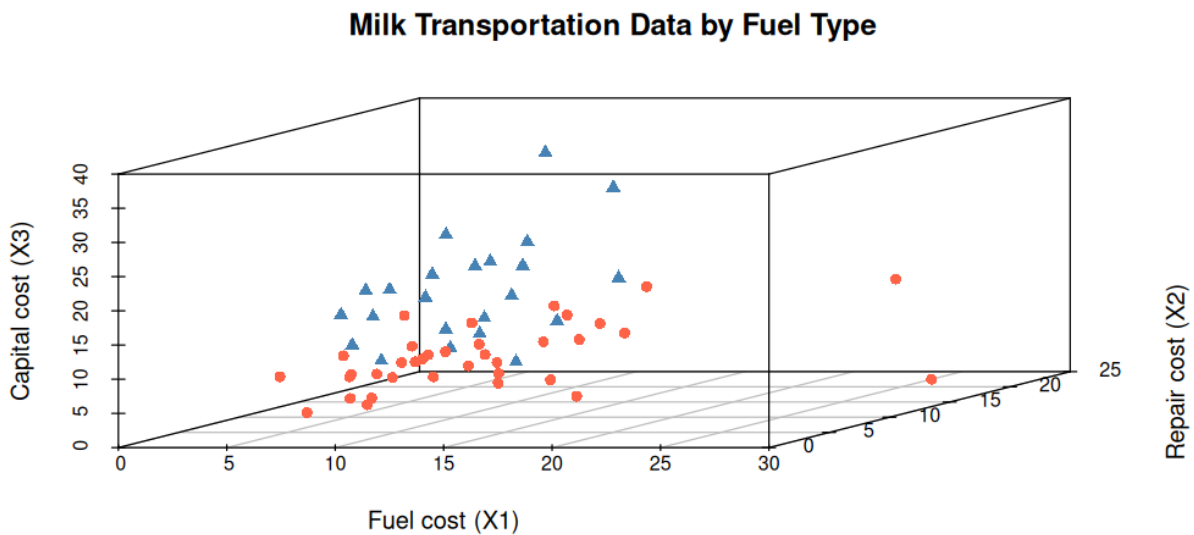


Figure 2.5: Three-dimensional scatterplot of the milk transportation data by fuel type.

The pairwise plots show that the two fuel types occupy somewhat different regions of the three-dimensional cost space, although the separation is not complete. In particular, the diesel group tends to lie at higher values of capital cost X_3 , while gasoline observations are more concentrated at lower X_3 values. This agrees with the univariate plots, where X_3 showed the clearest visual distinction between the two groups.

For the pairs (X_1, X_2) and (X_2, X_3) , a mild positive association is visible overall, though the pattern is not equally strong within each group. The pair (X_1, X_3) appears to show weaker linear structure. Thus, the dependence pattern seems to vary across variable pairs and possibly across fuel types.

The three-dimensional plot reinforces the same broad picture. The two groups overlap in the joint space, but they are not completely intermixed. Much of the visible separation appears to occur along the capital-cost direction, with repair cost contributing additional distinction in some regions. At the same time, a few points appear relatively isolated from the main clouds, suggesting the presence of unusual multivariate observations.

Overall, the multivariate plots indicate that the data contain both group separation and substantial overlap. They also suggest that the joint dependence structure is nontrivial and that some observations may deserve closer examination in later sections.

2.5 Correlation and Dependence Structure

To study linear dependence among the three cost variables, Table 2.5 reports the sample correlation matrices for the full dataset and for the gasoline and diesel groups separately. The same information is displayed visually in Figure 2.6.

Pair	Overall	Gasoline	Diesel
(X_1, X_2)	0.334	0.615	0.072
(X_1, X_3)	-0.064	0.162	0.166
(X_2, X_3)	0.363	0.305	0.221

Table 2.5: Pairwise sample correlations for the full dataset and the two fuel-type groups.

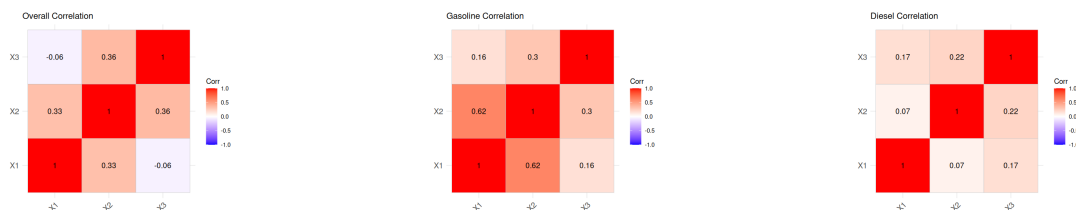


Figure 2.6: Correlation heatmaps for the full dataset, gasoline group, and diesel group.

Several features are worth noting. The pair (X_1, X_2) shows a moderate positive correlation overall, but this is driven mainly by the gasoline group, where the association is fairly strong; in the diesel group, the same pair is only weakly related. The pair (X_2, X_3) shows a positive association both overall and within each group, though the strength is only mild to moderate. In contrast, the overall correlation between X_1 and X_3 is close to zero, while within each group it is slightly positive.

Thus, the pooled correlation structure does not fully reflect the groupwise dependence patterns. In particular, the relation between fuel cost and repair cost appears to differ

substantially across fuel types. This suggests that part of the dependence structure is group-specific, and that combined-data summaries may obscure features visible at the subgroup level.

2.6 Initial Structural Insights

The exploratory analysis suggests that the gasoline and diesel groups differ in their overall cost profiles. Gasoline observations tend to be higher in fuel cost X_1 , whereas diesel observations tend to be higher in repair cost X_2 and, more noticeably, in capital cost X_3 . Among the three variables, capital cost appears to provide the clearest visual separation between the two groups.

At the same time, the two groups are not cleanly separated. The univariate and multivariate plots show substantial overlap, especially in fuel and repair cost, so any later classification problem is unlikely to be trivial. The data therefore appear to contain both group differences and high variance within groups, which may pose challenges for inference and classification.

The dependence structure also appears to vary by fuel type. In particular, the association between fuel cost and repair cost is much stronger in the gasoline group than in the diesel group, while the other pairwise relationships are comparatively weak to moderate. Thus, the joint structure of the data is not captured fully by pooled summaries alone.

Overall, the exploratory evidence points to a dataset with meaningful group differences, moderate overlap, and non-uniform dependence patterns across variables and groups. These features motivate a more careful examination of unusual observations and distributional assumptions in the next section and the following chapter.

2.7 Early Evidence of Outliers and Skewness

The exploratory plots suggest that a small number of observations may be unusually extreme, particularly in fuel cost X_1 . This is most visible in the gasoline group, where a few points lie well above the main body of the data. Capital cost X_3 also shows some unusually large diesel observations.

The marginal plots further suggest mild to moderate right-skewness in some variables, especially X_1 and X_3 . These features are important because they may affect both distributional diagnostics and later multivariate procedures.

The presence of outliers and skewness motivates a more detailed examination of these issues in the next chapter. In particular, it will be important to determine whether the observed extreme points are influential in a multivariate sense, whether they represent data errors or valid observations, and whether transformations or robust methods may be needed to mitigate their effects on inference and classification.

Chapter 3

Assumption Checking and Data Diagnostics

3.1 Purpose of the Diagnostic Stage

Before applying inferential and classification methods, it is necessary to examine whether the milk transportation data satisfy the assumptions required by later multivariate procedures. In particular, this chapter focuses on the distributional structure of the three cost variables X_1 , X_2 , and X_3 within the two fuel-type groups: Gasoline and Diesel.

The main diagnostic questions are:

- whether the variables exhibit unusual skewness, tail behavior, or extreme observations,
- whether the data are approximately normal at the univariate and multivariate levels,
- whether suspicious observations materially affect later analysis,
- whether transformation or selective outlier removal is necessary.

A key point is that, for most group-comparison and classification methods used later, *groupwise* multivariate normality is more relevant than normality of the pooled dataset. Combined-data normality is only needed for a smaller set of procedures, such as certain simultaneous confidence regions and some PCA-related inference. Therefore, diagnostics in this chapter are used to guide a *task-dependent* preprocessing decision rather than force a single preprocessing rule for all subsequent analyses.

3.2 Detection of Suspicious Observations

To identify potentially influential observations, outlier detection was carried out on the raw data using both univariate and multivariate criteria. At the univariate level, boxplot rules were applied within each fuel-type group. At the multivariate level, squared Mahalanobis distances were computed separately within the gasoline and diesel groups and compared against the corresponding chi-square reference pattern.

The boxplot rule flags two gasoline observations repeatedly: rows 9 and 21. In the diesel group, rows 47 and 52 appear unusual under at least one marginal distribution. Mahalanobis-distance based detection leads to a sharper conclusion: rows 9 and 21 are again clearly flagged in the gasoline group, while row 52 is the main multivariate outlier in the diesel group. Thus, rows 9 and 21 are the most consistently suspicious observations in the dataset, and both are associated primarily with unusually large values of X_1 .

Table 3.1: Observations flagged by boxplot and Mahalanobis rules

Row ID	FuelType	X_1	X_2	X_3	Flagged by
9	Gasoline	29.11	15.09	3.28	Boxplot, Mahalanobis
21	Gasoline	26.16	17.44	16.89	Boxplot, Mahalanobis
47	Diesel	9.77	17.86	35.18	Boxplot only
52	Diesel	15.90	12.90	19.09	Boxplot, Mahalanobis

Figure 3.1 shows the groupwise Mahalanobis Q–Q plots. The gasoline group exhibits two clear departures in the upper tail, corresponding to rows 9 and 21. The diesel group is more stable overall, though one upper-tail deviation remains visible. Figure 3.2 shows the same pattern in the (X_1, X_3) plane: the gasoline observations 9 and 21 lie far from the main data cloud and are visually more isolated than the diesel candidates.

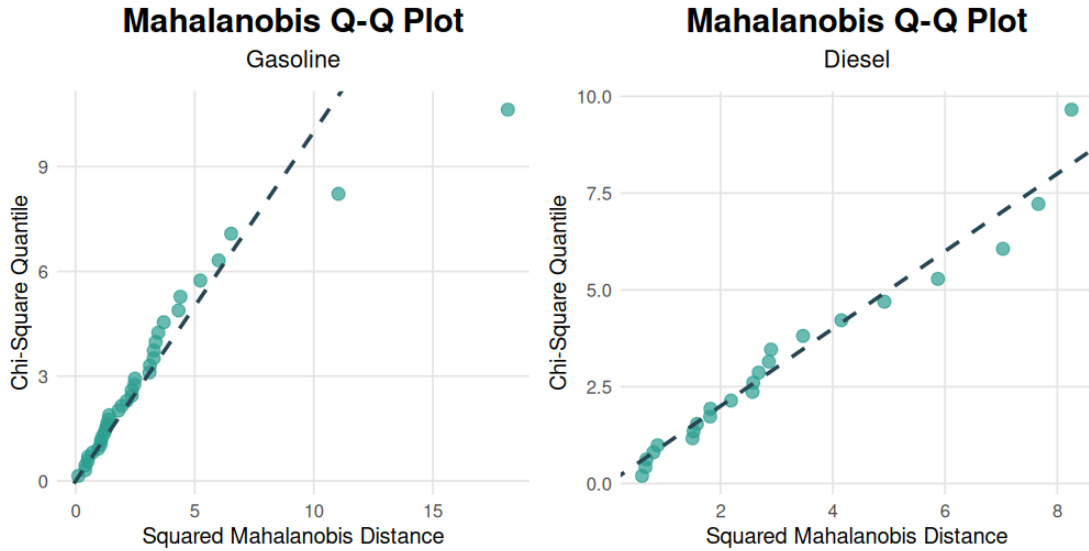


Figure 3.1: Groupwise Mahalanobis Q–Q plots for the gasoline and diesel groups.

The overall and groupwise outlier displays in Figures 3.2 and 3.3 reinforce the same conclusion: the dominant outlying behaviour is driven by a small number of observations, especially two gasoline units with unusually large fuel cost values.

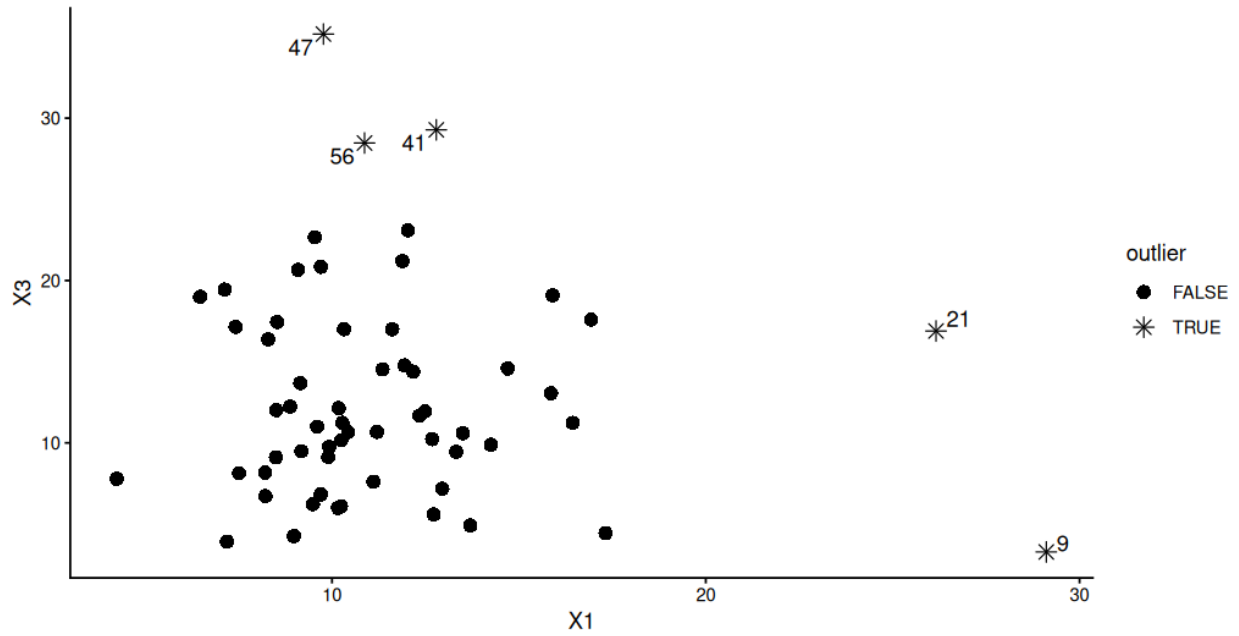


Figure 3.2: Overall outlier display.

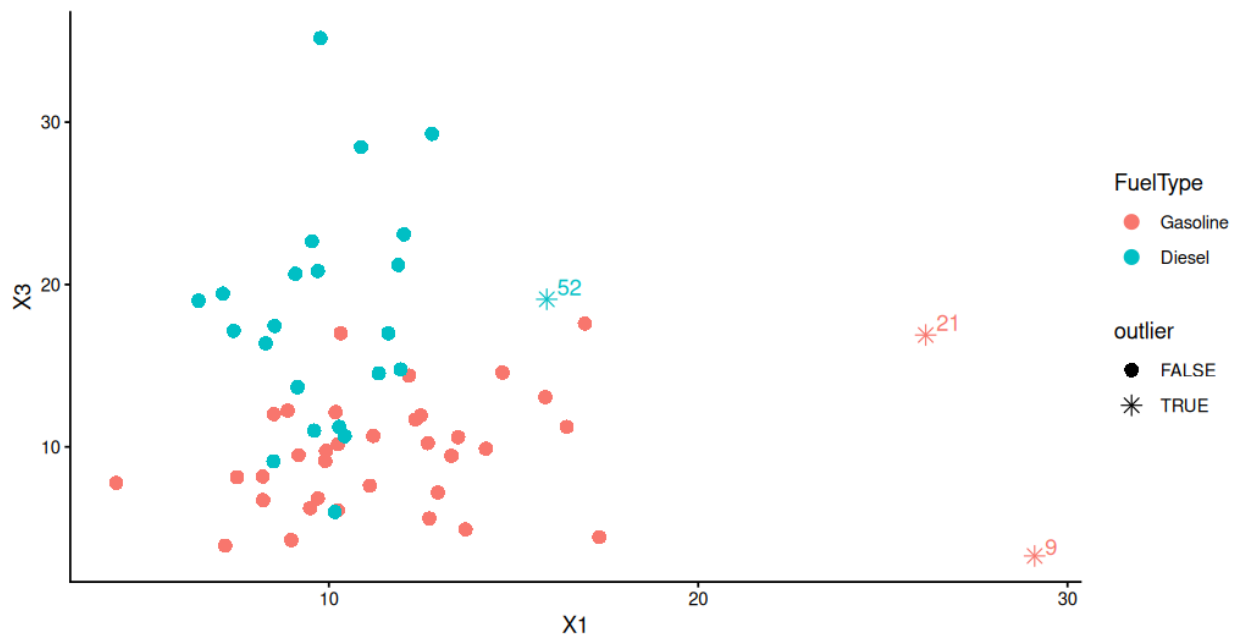


Figure 3.3: Groupwise outlier display.

At this stage, these observations are treated as *suspicious* rather than automatically deleted. Their effect on normality diagnostics and later methods is examined in the next sections before making any preprocessing decision.

3.3 Normality Assessment

Normality was assessed on the raw data at both the marginal and joint levels. Since later inference is primarily group-based, special attention is given to *groupwise* multivariate normality, while pooled-data normality is treated as a secondary diagnostic. The main question is whether the raw data may reasonably be modeled as two multivariate normal populations, or whether the departures from normality are too strong to ignore.

3.3.1 Univariate normality

Table 3.2 summarizes the three univariate normality tests applied to each variable, both for the pooled sample and within each fuel-type group. For the pooled data, X_1 and X_3 are rejected by all three tests, indicating clear marginal non-normality. Variable X_2 presents a weaker departure: Shapiro–Wilk and Anderson–Darling do not reject, but Lilliefors does, suggesting that its marginal behavior is less problematic than that of X_1 and X_3 .

Within groups, the main departure is concentrated in the gasoline sample. In particular, gasoline X_1 is rejected by all three tests, whereas gasoline X_2 and X_3 are accepted throughout. By contrast, none of the three diesel marginals is rejected by any of the tests considered. Thus, the marginal evidence points to a localized normality problem rather than a uniform failure across all variables and both groups.

Table 3.2: Univariate normality assessment for the raw data

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.8106	< 0.001 X	2.5898	< 0.001 X	0.1481	0.0025 X
Overall	X_2	0.9662	0.1002	0.6070	0.1096	0.1230	0.0267 X
Overall	X_3	0.9295	0.0021 X	1.0195	0.0102	0.1355	0.0088
Gasoline	X_1	0.8367	0.0001 X	1.6229	0.0003 X	0.1565	0.0259 X
Gasoline	X_2	0.9628	0.2623	0.4488	0.2628	0.1247	0.1677
Gasoline	X_3	0.9710	0.4532	0.2407	0.7572	0.0733	0.8946
Diesel	X_1	0.9623	0.5117	0.2518	0.7077	0.0915	0.8865
Diesel	X_2	0.9618	0.5000	0.3178	0.5153	0.1185	0.5496
Diesel	X_3	0.9687	0.6583	0.2861	0.5922	0.1112	0.6513

Overall, the strongest marginal departure occurs in gasoline fuel cost X_1 , which is fully consistent with the upper-tail anomalies and suspicious observations identified in Section 3.3.

3.3.2 Multivariate normality

Table 3.3 summarizes the multivariate normality diagnostics. The pooled sample is decisively non-normal: Mardia’s skewness and kurtosis, Henze–Zirkler, Royston, and Doornik–Hansen

all reject the null hypothesis. The gasoline group also fails the multivariate normality checks, with rejection under Mardia’s skewness and kurtosis, Royston’s test, Doornik–Hansen, and borderline rejection under Henze–Zirkler. In contrast, the diesel group is consistent with multivariate normality under all tests considered.

Table 3.3: Multivariate normality assessment for the raw data

Group	Mardia skewness		Mardia kurtosis		Henze–Zirkler		Royston		Doornik–Hansen	
	Stat.	p	Stat.	p	Stat.	p	Stat.	p	Stat.	p
Overall	66.347	< 0.001 $\mathbf{X}$	5.270	< 0.001 $\mathbf{X}$	1.251	0.001 $\mathbf{X}$	40.227	< 0.001 $\mathbf{X}$	23.437	< 0.001 $\mathbf{X}$
Gasoline	37.907	< 0.001 $\mathbf{X}$	2.780	0.005 $\mathbf{X}$	0.895	0.050 $\mathbf{X}$	18.487	< 0.001 $\mathbf{X}$	13.198	0.040 $\mathbf{X}$
Diesel	7.292	0.698	-0.430	0.667	0.543	0.583	1.713	0.634	6.084	0.414

These numerical results agree with the Mahalanobis Q–Q diagnostics discussed earlier. In particular, the raw pooled sample does not support a single multivariate normal model, and the gasoline group is the main source of multivariate departure. The diesel group, however, appears well behaved.

3.3.3 Interpretation

The raw-data normality assessment leads to three conclusions. First, the pooled dataset is clearly non-normal, both marginally and jointly. Second, this failure is driven mainly by the gasoline group, especially through the behavior of X_1 . Third, the diesel sample is broadly compatible with both univariate and multivariate normality. Hence the raw data are not well described by one common Gaussian population, but they may still be compatible with a *two-group* Gaussian structure after addressing a small number of influential observations.

For this reason, the next section examines whether the departures from normality persist after removing the suspicious observations identified in Section 3.3, before turning to transformation-based remedies.

3.4 Effect of Removing Suspicious Observations

The raw-data normality assessment suggested that the main departures from normality were driven by a small number of flagged observations, especially rows 9 and 21 in the gasoline group. To assess how strongly these observations influence the distributional conclusions, two deletion schemes were examined.

3.4.1 Candidate deletion rules

Two cleaned datasets were considered.

Table 3.4: Deletion schemes considered in the sensitivity analysis

Dataset	Deleted rows	Interpretation
<code>milk_no_out</code>	9, 21	targeted deletion of the two main gasoline outliers
<code>milk_nooutlier</code>	9, 21, 41, 47, 56	broadier deletion including additional diesel outliers

The two deletion schemes reflect different preprocessing goals. Removing rows 9 and 21 is a targeted adjustment based on the most consistently flagged observations in Section 3.3, whereas the broader deletion rule examines whether further diagnostic improvement is possible at the cost of removing additional data.

3.4.2 Results after removing rows 9 and 21

Table 3.5 reports the univariate normality diagnostics for the dataset obtained after deleting rows 9 and 21. The main change is at the group level: for both gasoline and diesel, all three variables are accepted by Shapiro–Wilk, Anderson–Darling, and Lilliefors. Hence the earlier marginal non-normality is no longer present within either group. At the pooled level, however, the conclusion is weaker. Variables X_1 and X_2 are acceptable under Shapiro–Wilk and Anderson–Darling but are still rejected by Lilliefors, while X_3 continues to show evidence against normality under all three tests.

Table 3.5: Univariate normality diagnostics after removing rows 9 and 21

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.9733	0.2381	0.5765	0.1281	0.1192	0.0425✗
Overall	X_2	0.9648	0.0959	0.5997	0.1141	0.1241	0.0290✗
Overall	X_3	0.9203	0.0011✗	1.1615	0.0045✗	0.1450	0.0044✗
Gasoline	X_1	0.9778	0.7036	0.3104	0.5374	0.1293	0.1585
Gasoline	X_2	0.9548	0.1712	0.4444	0.2684	0.1233	0.2111
Gasoline	X_3	0.9727	0.5410	0.2131	0.8401	0.0733	0.9142
Diesel	X_1	0.9623	0.5117	0.2518	0.7077	0.0915	0.8865
Diesel	X_2	0.9618	0.5000	0.3178	0.5153	0.1185	0.5496
Diesel	X_3	0.9687	0.6583	0.2861	0.5922	0.1112	0.6513

The multivariate results are summarized in Table 3.7. Here the improvement is sharper. For both gasoline and diesel, Mardia’s skewness and kurtosis, Henze–Zirkler, Royston, and Doornik–Hansen all accept multivariate normality. Thus removing rows 9 and 21 is sufficient to restore strong *groupwise* multivariate normality. In contrast, the pooled sample still shows some residual departure from joint normality: Mardia skewness, Henze–Zirkler, Royston, and Doornik–Hansen reject, while only Mardia kurtosis does not.

Therefore, the effect of removing rows 9 and 21 is highly structured. It largely resolves the normality problem within each fuel-type group, but it does not make the pooled sample fully normal. This supports the interpretation that the data are better viewed as two approximately normal populations, rather than one homogeneous normal sample.

3.4.3 Results after removing rows 9, 21, 41, 47, and 56

Table 3.6 reports the univariate normality diagnostics for the broader deletion rule. The groupwise conclusions remain favorable: for both gasoline and diesel, all three variables are accepted by Shapiro–Wilk, Anderson–Darling, and Lilliefors. Thus the broader deletion also yields satisfactory marginal behavior within each fuel-type group.

At the pooled level, the univariate diagnostics improve relative to the raw data, but the evidence is still mixed. Shapiro–Wilk and Anderson–Darling accept all three variables at the 5% level, whereas Lilliefors still rejects X_1 and X_2 . Hence pooled marginal normality is improved, though not uniformly across all tests.

Table 3.6: Univariate normality diagnostics after removing rows 9, 21, 41, 47, and 56

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	p -value	Statistic	p -value	Statistic	p -value
Overall	X_1	0.9716	0.2259	0.5893	0.1183	0.1281	0.0273✗
Overall	X_2	0.9659	0.1263	0.6326	0.0941	0.1236	0.0388✗
Overall	X_3	0.9577	0.0541	0.6790	0.0720	0.1146	0.0741
Gasoline	X_1	0.9778	0.7036	0.3104	0.5374	0.1293	0.1585
Gasoline	X_2	0.9548	0.1712	0.4444	0.2684	0.1233	0.2111
Gasoline	X_3	0.9727	0.5410	0.2131	0.8401	0.0733	0.9142
Diesel	X_1	0.9432	0.2759	0.3310	0.4846	0.1163	0.6804
Diesel	X_2	0.9530	0.4143	0.3877	0.3540	0.1405	0.3779
Diesel	X_3	0.9571	0.4875	0.3119	0.5224	0.1178	0.6609

At the multivariate level, the broader deletion also improves the pooled distributional behavior. According to Table 3.7, Mardia skewness, Mardia kurtosis, Royston, and Doornik–Hansen all accept multivariate normality for the pooled sample, while only Henze–Zirkler continues to reject. For both gasoline and diesel, all reported multivariate procedures accept normality. Thus the broader deletion yields strong groupwise support and somewhat stronger pooled support than the two-row deletion.

Overall, this broader deletion rule further stabilizes the diagnostic picture, especially for the pooled sample. However, it does so at the cost of removing additional diesel observations beyond the two gasoline points that were most consistently identified earlier.

Table 3.7: Multivariate normality after deleting suspicious observations

Dataset	Group	Mardia skewness		Mardia kurtosis		Henze–Zirkler		Royston		Doornik–Hansen	
		Stat.	<i>p</i>	Stat.	<i>p</i>	Stat.	<i>p</i>	Stat.	<i>p</i>	Stat.	<i>p</i>
<code>milk_no_out</code>	Overall	18.637	0.045✗	0.377	0.706	1.018	0.022✗	15.055	0.002✗	14.350	0.026✗
<code>milk_no_out</code>	Gasoline	6.379	0.783	-0.583	0.560	0.584	0.576	2.412	0.496	6.428	0.377
<code>milk_no_out</code>	Diesel	7.292	0.698	-0.430	0.667	0.543	0.583	1.713	0.634	6.084	0.414
<code>milk_nooutlier</code>	Overall	6.743	0.749	-1.703	0.089	0.995	0.026✗	7.525	0.057	9.611	0.142
<code>milk_nooutlier</code>	Gasoline	6.379	0.783	-0.583	0.560	0.584	0.576	2.412	0.496	6.428	0.377
<code>milk_nooutlier</code>	Diesel	7.864	0.642	-1.062	0.288	0.550	0.522	3.018	0.389	3.299	0.770

3.4.4 Comparison of the two deletion strategies

The comparison highlights a tradeoff rather than a single dominant choice. Removing rows 9 and 21 is sufficient to restore strong *groupwise* multivariate normality, which is the more relevant condition for many later group-based procedures. The broader deletion improves pooled normality somewhat further, but does so by removing three additional diesel observations and therefore entails greater loss of data.

The same tradeoff appears in classification performance. Table 3.8 shows that both deletion schemes reduce LOOCV QDA accuracy relative to the full raw data. Although the broader deletion performs slightly better than the two-row deletion, neither matches the accuracy achieved without deletion.

Table 3.8: LOOCV QDA accuracy under different deletion rules

Dataset	LOOCV accuracy
Raw data	0.8644
<code>milk_no_out</code>	0.8070
<code>milk_nooutlier</code>	0.8148

3.4.5 Implication for later analysis

The main failure of raw-data normality is driven by a small number of influential observations, especially rows 9 and 21, and once these are removed the data are reasonably modeled as two approximately multivariate normal populations. At the same time, deleting suspicious observations weakens classification performance, so any removal should be treated as method-dependent rather than universal.

3.5 Effect of Transformations

3.5.1 Aim of the transformation analysis

The previous sections showed that the raw data fail normality primarily because of strong upper-tail behavior in the gasoline group, especially in X_1 . This section examines whether

such departures may be reduced *without deleting observations*, by applying standard monotone transformations to the full dataset. The purpose is to isolate the effect of transformation alone, separate from the effect of outlier removal.

3.5.2 Transformations considered

Three transformations were considered.

Table 3.9: Transformations considered in the analysis

Transformation	Definition
Logarithmic	$x^* = \log x$
Box-Cox	$x^* = \begin{cases} \frac{x^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log x, & \lambda = 0 \end{cases}$
Yeo-Johnson	$x^* = \begin{cases} \frac{(x+1)^\lambda - 1}{\lambda}, & x \geq 0, \lambda \neq 0, \\ \log(x+1), & x \geq 0, \lambda = 0, \\ -\frac{(1-x)^{2-\lambda} - 1}{2-\lambda}, & x < 0, \lambda \neq 2, \\ -\log(1-x), & x < 0, \lambda = 2 \end{cases}$

All three transformations were applied componentwise to the variables X_1, X_2, X_3 . Since the data are strictly positive, the main motivation is the reduction of skewness and tail-heaviness, especially in fuel cost X_1 .

3.5.3 Effect on univariate normality

Tables 3.10, 3.11, and 3.12 summarize the univariate normality diagnostics for the transformed datasets.

The groupwise conclusions are uniformly favorable: under all three transformations, the gasoline and diesel marginals are accepted by Shapiro-Wilk, Anderson-Darling, and Lilliefors for all three variables. Thus, from a purely marginal viewpoint, each transformation improves the within-group behavior relative to the raw data.

At the pooled level, however, the three transformations behave differently. The logarithmic transformation still leaves clear rejection for X_1 and X_2 under some tests. Box-Cox improves X_2 and X_3 substantially, but X_1 is still rejected by Shapiro-Wilk. Yeo-Johnson gives the cleanest pooled marginal picture, with no rejection at the 5% level under any of the three reported tests for any variable.

Table 3.10: Univariate normality diagnostics for the log-transformed data

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.9529	0.0231✗	0.7417	0.0505	0.0911	0.2591
Overall	X_2	0.9523	0.0216✗	0.8989	0.0204✗	0.1085	0.0812
Overall	X_3	0.9893	0.8842	0.2083	0.8586	0.0639	0.7951
Gasoline	X_1	0.9557	0.1580	0.5202	0.1739	0.0899	0.6527
Gasoline	X_2	0.9485	0.0940	0.5396	0.1551	0.0977	0.5195
Gasoline	X_3	0.9652	0.3087	0.4005	0.3444	0.1180	0.2323
Diesel	X_1	0.9852	0.9734	0.1838	0.8985	0.0840	0.9434
Diesel	X_2	0.9255	0.0873	0.7167	0.0530	0.1740	0.0690
Diesel	X_3	0.9685	0.6522	0.3180	0.5149	0.1215	0.5088

Table 3.11: Univariate normality diagnostics for the Box–Cox transformed data

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.9591	0.0452✗	0.6276	0.0974	0.0835	0.3870
Overall	X_2	0.9800	0.4400	0.4431	0.2775	0.0937	0.2220
Overall	X_3	0.9925	0.9751	0.1426	0.9696	0.0596	0.8674
Gasoline	X_1	0.9556	0.1567	0.4773	0.2233	0.0889	0.6699
Gasoline	X_2	0.9774	0.6585	0.2633	0.6804	0.0901	0.6496
Gasoline	X_3	0.9752	0.5842	0.2700	0.6574	0.1018	0.4530
Diesel	X_1	0.9847	0.9684	0.1958	0.8775	0.0859	0.9310
Diesel	X_2	0.9587	0.4378	0.3984	0.3378	0.1358	0.3307
Diesel	X_3	0.9812	0.9262	0.2388	0.7526	0.1043	0.7434

Table 3.12: Univariate normality diagnostics for the Yeo–Johnson transformed data

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.9615	0.0592	0.5928	0.1189	0.0820	0.4174
Overall	X_2	0.9789	0.3942	0.4591	0.2537	0.0961	0.1921
Overall	X_3	0.9921	0.9673	0.1452	0.9666	0.0597	0.8658
Gasoline	X_1	0.9576	0.1808	0.4543	0.2547	0.0871	0.7004
Gasoline	X_2	0.9764	0.6225	0.2731	0.6469	0.0904	0.6447
Gasoline	X_3	0.9757	0.5990	0.2611	0.6881	0.1002	0.4789
Diesel	X_1	0.9847	0.9689	0.1952	0.8788	0.0860	0.9304
Diesel	X_2	0.9582	0.4290	0.4006	0.3336	0.1358	0.3312
Diesel	X_3	0.9807	0.9183	0.2416	0.7430	0.1055	0.7285

3.5.4 Effect on multivariate normality

Table 3.13 summarizes the multivariate normality diagnostics for the transformed datasets.

The multivariate picture is more mixed than the univariate one. The logarithmic transformation does not restore multivariate normality consistently: the pooled sample is still rejected by Mardia skewness and Royston, and the gasoline group is rejected by Mardia skewness and Doornik–Hansen, although the diesel group is accepted by all reported tests. Box–Cox improves the diagnostics further, but still yields rejection by Mardia skewness in the gasoline group and by Doornik–Hansen in the pooled and diesel samples. Yeo–Johnson behaves similarly: it gives strong pooled and groupwise acceptance under Mardia kurtosis, Henze–Zirkler, and Royston, but Mardia skewness still rejects for gasoline and Doornik–Hansen rejects for the pooled and diesel samples.

Hence no transformation yields uniform acceptance across all multivariate procedures. The main remaining difficulties are concentrated in gasoline skewness and in Doornik–Hansen rejections for some transformed datasets.

Table 3.13: Multivariate normality diagnostics for the transformed datasets

Dataset	Group	Mardia skewness		Mardia kurtosis		Henze–Zirkler		Royston		Doornik–Hansen	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Log	Overall	20.499	0.025✗	1.402	0.161	0.730	0.324	12.662	0.005✗	7.628	0.267
Log	Gasoline	25.416	0.005✗	1.653	0.098	0.805	0.118	7.487	0.059	20.660	0.002✗
Log	Diesel	7.556	0.672	-0.905	0.365	0.534	0.612	3.390	0.335	5.215	0.517
Box–Cox	Overall	13.377	0.203	0.911	0.362	0.752	0.275	6.861	0.077	18.372	0.005✗
Box–Cox	Gasoline	21.849	0.016✗	1.168	0.243	0.732	0.222	4.219	0.242	9.270	0.159
Box–Cox	Diesel	6.158	0.802	-1.026	0.305	0.491	0.735	0.671	0.880	20.738	0.002✗
Yeo–Johnson	Overall	12.908	0.229	0.788	0.431	0.761	0.256	6.463	0.092	28.737	< 0.001✗
Yeo–Johnson	Gasoline	21.348	0.019✗	1.068	0.286	0.732	0.222	3.953	0.270	11.833	0.066
Yeo–Johnson	Diesel	6.214	0.797	-1.035	0.301	0.494	0.728	0.698	0.874	17.738	0.007

3.5.5 Comparison across transformations

Tables 3.10–3.12 and Table 3.13 allow a direct comparison of the three transformation strategies. At the univariate level, all three transformations improve the within-group marginals substantially, and each produces acceptable groupwise behavior for gasoline and diesel. Among the three, Yeo–Johnson gives the cleanest pooled marginal picture, while Box–Cox also performs well overall. The logarithmic transformation is somewhat weaker, since pooled departures remain more visible for X_1 and X_2 .

At the multivariate level, however, the conclusions are less uniform. Log transformation does not resolve the main joint-normality issues and continues to show rejection for the pooled sample and for the gasoline group under some procedures. Box–Cox and Yeo–Johnson perform better overall, but neither yields consistent acceptance across all reported multivariate tests. In particular, the remaining difficulties are concentrated in gasoline skewness and in Doornik–Hansen rejections for some transformed datasets.

Table 3.14: Comparison of Box–Cox and Yeo–Johnson transformations on the full dataset

Transformation	LOOCV QDA accuracy	Gasoline error	Diesel error
Box–Cox	0.8475	0.0833	0.2609
Yeo–Johnson	0.8475	0.0833	0.2609

Table 3.14 shows that Box–Cox and Yeo–Johnson are indistinguishable from the standpoint of QDA performance, both yielding the same overall accuracy and the same class-wise error rates. The choice between them therefore rests on the diagnostic results. Since Yeo–Johnson gives the cleaner marginal normality picture, while offering no disadvantage relative to Box–Cox in classification, it would be the preferred transformation if all observations had to be retained.

3.5.6 Recovering overall normality

Although neither transformation alone nor targeted deletion alone yields uniformly strong evidence for pooled multivariate normality, the combination of the two does. In particular, when rows 9 and 21 are removed and a Yeo–Johnson transformation is applied, the pooled sample is accepted by all reported univariate and multivariate normality procedures, and the same is true for the gasoline and diesel groups separately.

This result shows that overall normality can indeed be recovered, but only after heavy preprocessing. Since this combined strategy changes the measurement scale, removes observations, and still reduces classification accuracy relative to the raw data, it is less preferable than the earlier methods. However, it does provide a useful benchmark for the best-case normality scenario, and it confirms that the main departures from normality are driven by a small number of influential observations and by skewness in the gasoline group.

3.6 Regression-Based Imputation of Suspicious Observations

As an alternative to deleting suspicious observations, a regression-based imputation strategy was also examined. The idea was to retain all observations, but replace suspicious values of X_1 and X_3 by fitted values obtained from a multivariate multiple regression on X_2 , estimated from the corresponding cleaned sample. This was done on the raw, log, Box–Cox, and Yeo–Johnson scales.

Table 3.15: Regression models used for imputation

Scale	$X_1 \sim X_2$		$X_3 \sim X_2$		MANOVA (Pillai)	
	Slope	<i>p</i> -value	Slope	<i>p</i> -value	Statistic	<i>p</i> -value
Raw	0.3616	0.0069	0.3029	0.0564	0.2466	0.0124
Log	0.2306	0.0018	0.2154	0.0554	0.2877	0.0052
Box–Cox	0.0564	0.0009	0.1409	0.0297	0.3260	0.0022
Yeo–Johnson	0.0433	0.0009	0.1299	0.0291	0.3264	0.0022

Table 3.15 shows that the regression relationship used for imputation is strongest for X_1 and remains statistically meaningful across all four scales. The fitted values were then used to replace the suspicious observations before reassessing normality.

3.6.1 Univariate normality after imputation

Tables 3.16–3.19 summarize the univariate diagnostics after imputation.

Raw-scale imputation improves the groupwise marginals substantially: all three tests accept all gasoline and diesel variables. At the pooled level, however, Lilliefors still rejects X_1 and X_2 .

Log-scale imputation yields acceptable groupwise marginals throughout, and also fixes the pooled behavior of X_1 and X_3 . The only remaining pooled univariate difficulty is X_2 , which is still rejected by Shapiro–Wilk and Anderson–Darling.

Box–Cox and Yeo–Johnson imputation give the cleanest univariate results. Under both, all three pooled marginals are accepted by all three tests, and the same is true within both fuel-type groups.

Table 3.16: Univariate normality diagnostics after raw-scale imputation

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.9736	0.2266	0.6210	0.1011	0.1321	0.0121✗
Overall	X_2	0.9662	0.1002	0.6070	0.1096	0.1230	0.0267✗
Overall	X_3	0.9642	0.0802	0.5877	0.1200	0.0933	0.2271
Gasoline	X_1	0.9812	0.7835	0.2728	0.6480	0.1230	0.1827
Gasoline	X_2	0.9628	0.2623	0.4488	0.2628	0.1247	0.1677
Gasoline	X_3	0.9756	0.5971	0.2174	0.8290	0.0653	0.9614
Diesel	X_1	0.9231	0.0778	0.6387	0.0839	0.1556	0.1579
Diesel	X_2	0.9618	0.5000	0.3178	0.5153	0.1185	0.5496
Diesel	X_3	0.9694	0.6751	0.2303	0.7804	0.1060	0.7209

Table 3.17: Univariate normality diagnostics after log-scale imputation

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.9728	0.2078	0.3162	0.5319	0.0901	0.2746
Overall	X_2	0.9523	0.0216✗	0.8989	0.0204✗	0.1085	0.0812
Overall	X_3	0.9892	0.8817	0.2005	0.8771	0.0627	0.8163
Gasoline	X_1	0.9670	0.3505	0.2697	0.6584	0.1050	0.4028
Gasoline	X_2	0.9485	0.0940	0.5396	0.1551	0.0977	0.5195
Gasoline	X_3	0.9641	0.2857	0.4146	0.3184	0.1141	0.2769
Diesel	X_1	0.9852	0.9734	0.1838	0.8985	0.0840	0.9434
Diesel	X_2	0.9255	0.0873	0.7167	0.0530	0.1740	0.0690
Diesel	X_3	0.9685	0.6522	0.3180	0.5149	0.1215	0.5088

Table 3.18: Univariate normality diagnostics after Box–Cox-scale imputation

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.9820	0.5283	0.2410	0.7633	0.0823	0.4116
Overall	X_2	0.9800	0.4400	0.4431	0.2775	0.0937	0.2220
Overall	X_3	0.9898	0.9035	0.1738	0.9229	0.0746	0.5710
Gasoline	X_1	0.9761	0.6145	0.2262	0.8032	0.0996	0.4881
Gasoline	X_2	0.9774	0.6585	0.2633	0.6804	0.0901	0.6496
Gasoline	X_3	0.9726	0.5012	0.2950	0.5779	0.0965	0.5395
Diesel	X_1	0.9847	0.9684	0.1958	0.8775	0.0859	0.9310
Diesel	X_2	0.9587	0.4378	0.3984	0.3378	0.1358	0.3307
Diesel	X_3	0.9812	0.9262	0.2388	0.7526	0.1043	0.7434

Table 3.19: Univariate normality diagnostics after Yeo–Johnson-scale imputation

Group	Variable	Shapiro–Wilk		Anderson–Darling		Lilliefors	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Overall	X_1	0.9832	0.5868	0.2320	0.7919	0.0815	0.4267
Overall	X_2	0.9789	0.3942	0.4591	0.2537	0.0961	0.1921
Overall	X_3	0.9895	0.8927	0.1754	0.9201	0.0746	0.5712
Gasoline	X_1	0.9774	0.6571	0.2220	0.8158	0.0992	0.4955
Gasoline	X_2	0.9764	0.6225	0.2731	0.6469	0.0904	0.6447
Gasoline	X_3	0.9730	0.5116	0.2880	0.5988	0.0950	0.5650
Diesel	X_1	0.9847	0.9689	0.1952	0.8788	0.0860	0.9304
Diesel	X_2	0.9582	0.4290	0.4006	0.3336	0.1358	0.3312
Diesel	X_3	0.9807	0.9183	0.2416	0.7430	0.1055	0.7285

3.6.2 Multivariate normality after imputation

Table 3.20 summarizes the multivariate results.

Raw-scale imputation is enough to restore multivariate normality within both gasoline and diesel, but the pooled sample is still rejected by Henze–Zirkler and Doornik–Hansen.

Log-scale imputation gives the strongest multivariate outcome. All five multivariate procedures accept the pooled sample, and all five also accept both fuel-type groups.

Box–Cox and Yeo–Johnson imputation also stabilize the data substantially. Under both, gasoline is accepted by all multivariate tests. However, Doornik–Hansen still rejects the

pooled sample and the diesel group, so the multivariate improvement is not as complete as under log-scale imputation.

Table 3.20: Multivariate normality diagnostics after imputation

Scale	Group	Mardia skewness		Mardia kurtosis		Henze–Zirkler		Royston		Doornik–Hansen	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Raw	Overall	6.736	0.750	-1.681	0.093	1.153	0.005	7.255	0.065	15.887	0.014✗
Raw	Gasoline	6.218	0.797	-0.639	0.523	0.606	0.526	1.631	0.659	9.296	0.158
Raw	Diesel	6.802	0.744	-1.392	0.164	0.682	0.229	3.568	0.315	2.921	0.819
Log	Overall	13.053	0.221	-0.286	0.775	0.678	0.456	7.692	0.053	2.942	0.816
Log	Gasoline	13.096	0.218	0.249	0.804	0.646	0.416	5.276	0.156	8.630	0.195
Log	Diesel	7.556	0.672	-0.905	0.365	0.534	0.612	3.390	0.335	5.215	0.517
Box–Cox	Overall	7.043	0.721	-0.781	0.435	0.767	0.245	1.425	0.701	20.532	0.002✗
Box–Cox	Gasoline	9.024	0.530	-0.423	0.672	0.616	0.497	1.076	0.790	9.320	0.156
Box–Cox	Diesel	6.158	0.802	-1.026	0.305	0.491	0.735	0.671	0.880	20.738	0.002
Yeo–Johnson	Overall	6.767	0.747	-0.849	0.396	0.781	0.219	1.381	0.711	29.117	< 0.001✗
Yeo–Johnson	Gasoline	8.845	0.547	-0.473	0.636	0.623	0.477	1.007	0.806	12.325	0.055
Yeo–Johnson	Diesel	6.214	0.797	-1.035	0.301	0.494	0.728	0.698	0.874	17.738	0.007✗

3.6.3 Conclusion

The imputation results show that model-based replacement of suspicious observations can repair the distributional structure without physically removing observations. Raw-scale imputation is enough to recover strong groupwise multivariate normality, but not full pooled normality. Box–Cox and Yeo–Johnson imputation give the cleanest univariate results, but they still leave Doornik–Hansen rejections at the multivariate level. If imputation were to be pursued as a preprocessing strategy, log-scale imputation would be the most attractive choice, since it is the only version that yields consistent multivariate acceptance for the pooled sample as well as for both fuel-type groups.

3.7 Final Preprocessing Decision

The results of Chapter 3 show that preprocessing should be chosen according to the objective of the analysis. The main conclusions are as follows.

- **Exploratory analysis and visualization:** use the *raw dataset*. This preserves the observed sample structure and presents the data as originally recorded.
- **Classification (LDA/QDA):** use the *raw dataset* when the goal is predictive classification. Although the raw data do not satisfy the normality assumptions perfectly, they yield the best empirical classification performance among the datasets considered.
- **Group-based parametric inference:** use the *raw dataset with rows 9 and 21 removed*. This version gives the strongest support for groupwise multivariate normality and is therefore the most appropriate choice for inference based on the gasoline–diesel group structure.

- **Procedures requiring pooled multivariate normality:** if a later method specifically requires approximate normality of the combined sample, then stronger preprocessing may be used, such as broader deletion or a transformation-based adjustment.
- **If the suspicious observations are known to be erroneous:** they should not be retained unchanged. In that case, either removal or model-based imputation would be appropriate, depending on whether preserving sample size is more important than preserving the original observed values.

Overall, the data are most reasonably interpreted as two approximately multivariate normal groups together with a small number of influential observations. The final dataset choice should therefore follow the statistical purpose of the method being used.

Chapter 4

Two-Sample Multivariate Inference for Gasoline and Diesel Fleets

4.1 Inferential Setup

This chapter develops the formal two-sample multivariate inference for the milk transportation data. The analysis is carried out on the dataset `milk_no_out_yj`, which was selected as the working inferential dataset because it provides the most defensible basis for applying classical multivariate procedures. In particular, it reflects the preprocessing choice made earlier for assumption-based inference rather than for classification.

The response vector consists of the three transportation-cost variables X_1 , X_2 , and X_3 , where X_1 denotes fuel cost per mile, X_2 denotes repair cost per mile, and X_3 denotes capital cost per mile. The two groups correspond to gasoline and diesel vehicles.

Let $\boldsymbol{\mu}_G$ and $\boldsymbol{\mu}_D$ denote the population mean vectors for the gasoline and diesel groups, and let $\boldsymbol{\Sigma}_G$ and $\boldsymbol{\Sigma}_D$ denote the corresponding covariance matrices. The main inferential questions in this chapter are whether the two groups have the same mean vector and whether their covariance matrices may be treated as equal.

The inferential dataset contains 34 gasoline observations and 23 diesel observations. On this basis, the chapter proceeds from descriptive multivariate summaries to formal tests of covariance homogeneity and equality of mean vectors, followed by simultaneous confidence intervals for componentwise mean differences.

4.2 Descriptive Multivariate Summaries

The gasoline group contains 34 observations, while the diesel group contains 23 observations.

Table 4.1: Group mean vectors and mean difference

Group/Contrast	X_1	X_2	X_3
Gasoline	4.8077	3.5278	2.3983
Diesel	4.4980	4.3995	3.0404
Gasoline – Diesel	0.3097	-0.8718	-0.6421

Table 4.1 shows that the gasoline group has a slightly larger mean for X_1 , whereas the diesel group has larger means for X_2 and X_3 . Thus, at the descriptive level, the two groups appear to differ more clearly in repair and capital cost than in fuel cost.

Table 4.2: Sample covariance matrix: Gasoline group

	X_1	X_2	X_3
X_1	0.6856	0.4433	0.0831
X_2	0.4433	1.4801	0.1383
X_3	0.0831	0.1383	0.1375

Table 4.3: Sample covariance matrix: Diesel group

	X_1	X_2	X_3
X_1	0.3449	0.0590	0.0339
X_2	0.0590	1.9568	0.0585
X_3	0.0339	0.0585	0.1725

4.3 Equality of Covariance Matrices

Before applying the pooled two-sample Hotelling T^2 procedure, it is necessary to examine whether the covariance matrices of the gasoline and diesel groups may reasonably be treated as equal. For this purpose, Box's M test was used to assess the null hypothesis that the two population covariance matrices are the same.

Table 4.4: Box's M test for homogeneity of covariance matrices

Test statistic	df	p -value	Decision at 5%
5.9798	6	0.4255	Do not reject equality

Table 4.4 shows that Box's M test does not reject the null hypothesis of equal covariance matrices. Thus, there is no statistically significant evidence that the gasoline and diesel groups differ in their covariance structure.

This conclusion is consistent with the descriptive covariance summaries from the previous section. Although some numerical differences are present, particularly in the covariance between X_1 and X_2 and in the marginal variances, these differences are not large enough to rule out a common covariance model. Accordingly, the assumption of covariance homogeneity appears reasonable for the inferential analysis carried out in this chapter.

On this basis, it is appropriate to proceed with the pooled-covariance two-sample Hotelling T^2 test for comparing the mean vectors of the two groups.

4.4 Equality of Mean Vectors

We now test whether the gasoline and diesel groups have the same mean vector. Under the pooled-covariance model justified in the previous section, the appropriate procedure is the two-sample Hotelling T^2 test for the null hypothesis $H_0 : \boldsymbol{\mu}_G = \boldsymbol{\mu}_D$.

Table 4.5: Two-sample Hotelling T^2 test for equality of mean vectors

T^2 statistic	F value	df ₁	df ₂	p -value
50.5678	16.243	3	53	1.3×10^{-7}

Table 4.5 shows overwhelming evidence against the null hypothesis of equal mean vectors. Thus, the gasoline and diesel groups differ significantly in their joint mean transportation-cost profile.

This conclusion is consistent with the descriptive summaries in Section 2. The gasoline group has a somewhat larger mean for X_1 , whereas the diesel group has larger means for X_2 and X_3 . Although each coordinate need not differ significantly on its own, the Hotelling procedure shows that the three variables jointly produce a clear multivariate location difference between the two fleets.

Therefore, the inferential evidence supports the conclusion that gasoline and diesel vehicles do not have the same mean cost structure in the transformed inferential dataset.

4.5 Simultaneous Confidence Intervals for Mean Differences

To identify which variables contribute most clearly to the overall multivariate difference, simultaneous confidence intervals were constructed for the componentwise mean contrasts. These intervals correspond to the contrast Gasoline – Diesel and are reported in Table 4.6 and Figure 4.1.

Table 4.6: Simultaneous confidence intervals for componentwise mean differences

Variable	Lower	Upper	Includes 0?
X_1	-0.2789	0.8982	Yes
X_2	-1.8982	0.1547	Yes
X_3	-0.9512	-0.3331	No

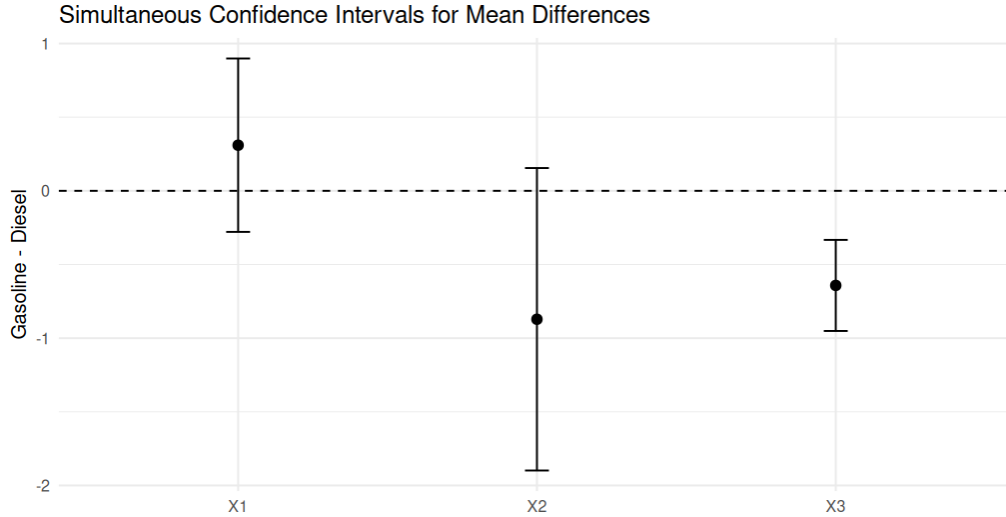


Figure 4.1: Simultaneous confidence intervals for the mean differences, Gasoline – Diesel.

The simultaneous intervals refine the conclusion of the Hotelling T^2 test by showing where the multivariate difference is most clearly concentrated. Among the three variables, only X_3 has an interval that excludes zero. Thus, at the simultaneous confidence level, X_3 is the only component for which there is clear evidence of a difference between the two group means.

By contrast, the intervals for X_1 and X_2 both contain zero. This means that, although these variables contribute to the joint multivariate separation, their individual mean differences are not large enough to be isolated after simultaneous adjustment.

The sign of the interval for X_3 is negative throughout, and since the contrast is Gasoline – Diesel, this indicates that the diesel group has the larger mean for X_3 . Therefore, the strongest componentwise evidence of group separation comes from capital cost, while fuel cost and repair cost do not show individually significant differences at the same simultaneous level.

4.6 Graphical Synthesis of Group Differences

Figures 4.2 and 4.3 give a visual summary of the multivariate difference between the gasoline and diesel groups.

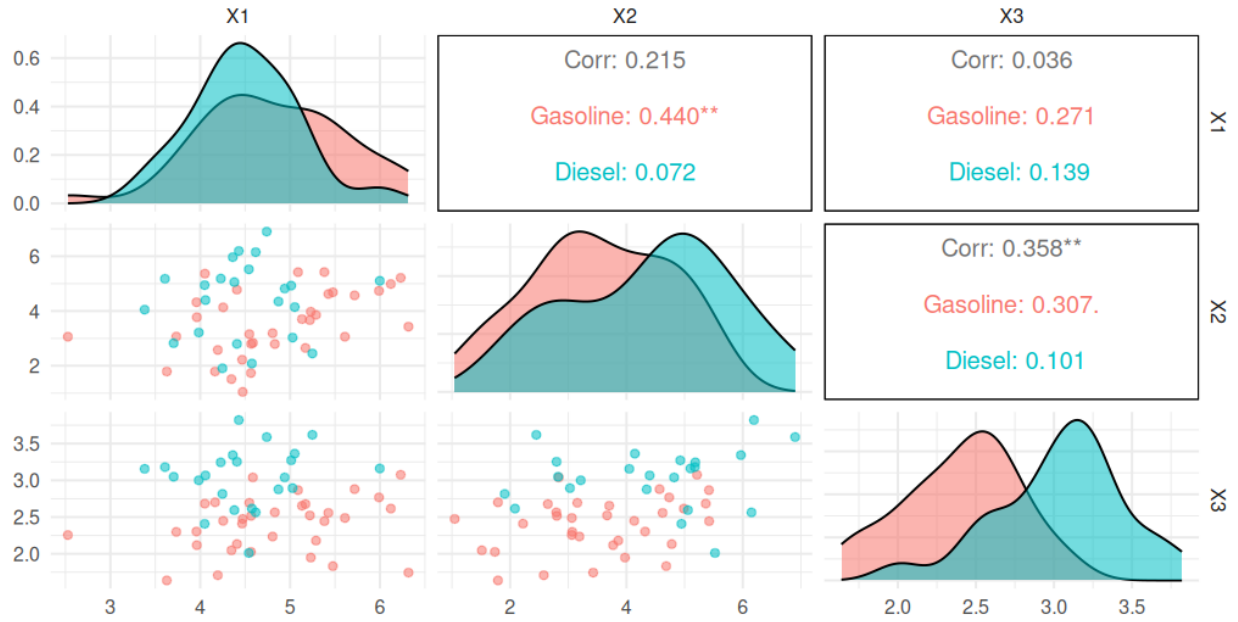


Figure 4.2: Scatterplot matrix and marginal densities for the inferential dataset.

Figure 4.2 shows substantial overlap between the two groups in all pairwise projections, so the separation is not complete. However, the marginal distribution of X_3 shows the clearest shift, with diesel tending to take larger values than gasoline. This agrees with the simultaneous confidence intervals, where only X_3 excluded zero. By contrast, the overlap in X_1 and X_2 is stronger, which is consistent with their intervals containing zero.

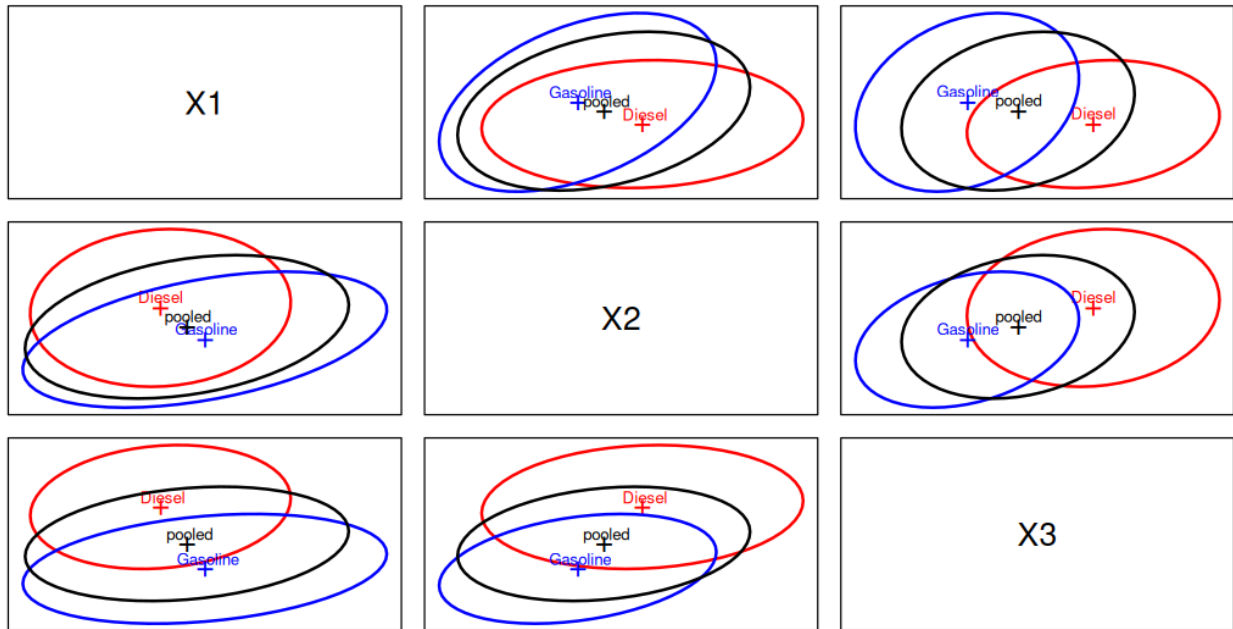


Figure 4.3: Pairwise covariance ellipses for the gasoline, diesel, and pooled samples.

Figure 4.3 conveys the same picture in terms of group centers and covariance structure. The ellipses overlap appreciably, but their centers are displaced, especially in panels involving X_3 . The ellipse shapes are not identical, yet they remain broadly similar, which is in line with the non-significant Box's M result.

Overall, the plots support the formal conclusions of the previous sections: the main difference between the two fleets is a shift in mean structure, not a strong difference in covariance structure, and X_3 appears to be the clearest source of separation.

Chapter 5

LDA and QDA

5.1 Objective and datasets considered

The aim of this chapter is to classify milk transportation units into the two fuel-type groups, *Gasoline* and *Diesel*, using the three cost variables

$$X_1 = \text{fuel cost}, \quad X_2 = \text{repair cost}, \quad X_3 = \text{capital cost},$$

all measured in cents per mile. We compare two classical discriminant methods, namely linear discriminant analysis (LDA) and quadratic discriminant analysis (QDA), and study how their classification performance changes under different preprocessing choices.

Since this chapter is prediction-oriented, the main basis of comparison is leave-one-out cross-validation (LOOCV), summarized through confusion matrices, overall accuracy, and class-wise error rates. Accordingly, the full raw dataset is treated as the primary dataset for classification, while transformed and outlier-removed versions are used only as sensitivity checks.

The four dataset versions considered are:

- **milk**: the full raw dataset. This is the primary dataset for classification comparison.
- **milk_yj**: the Yeo–Johnson transformed full dataset. This is used to assess whether transformation improves classification performance.
- **milk_no_out**: the raw dataset after removing the two influential gasoline outliers. This is used to assess the effect of outlier removal on classification.
- **milk_no_out_yj**: the Yeo–Johnson transformed dataset after outlier removal. This is used mainly as a sensitivity-analysis dataset and also for cleaner visualization of decision boundaries.

The chapter proceeds by first comparing LDA across these four dataset versions using LOOCV. The corresponding linear decision boundaries are then examined. Next, the same comparison is carried out for QDA, followed by a study of the quadratic decision boundaries. Finally, LDA and QDA are compared directly in order to determine the most suitable classification rule for this dataset.

5.2 LDA: cross-validated classification performance

We first compare linear discriminant analysis (LDA) across the four dataset versions using leave-one-out cross-validation. Since LDA assumes a common covariance matrix across the two groups, it produces a linear classification rule. For the present study, its performance is evaluated through the LOOCV confusion matrix, overall accuracy, and class-wise misclassification rates.

Table 5.1: LOOCV performance of LDA under different preprocessing choices

Dataset	GG	GD	DG	DD	Accuracy	Gasoline error	Diesel error
<code>milk</code>	33	3	8	15	0.8136	0.0833	0.3478
<code>milk_yj</code>	31	5	7	16	0.7966	0.1389	0.3043
<code>milk_no_out</code>	31	3	9	14	0.7895	0.0882	0.3913
<code>milk_no_out_yj</code>	29	5	7	16	0.7895	0.1471	0.3043

GG = Gasoline classified as Gasoline, GD = Gasoline classified as Diesel, DG = Diesel classified as Gasoline, DD = Diesel classified as Diesel.

Among the four versions, the full raw dataset `milk` gives the highest LOOCV accuracy. The transformed dataset `milk_yj` slightly reduces overall accuracy, although it lowers diesel misclassification relative to the raw-data LDA fit. Removing the two influential gasoline observations does not improve LDA performance; in fact, `milk_no_out` gives the largest diesel error rate. The combined preprocessing strategy `milk_no_out_yj` also fails to outperform the raw-data fit.

Overall, the LDA results show that preprocessing does not improve predictive performance in this dataset. Gasoline is classified quite well in all four cases, whereas diesel is consistently more difficult to classify. Thus, within the LDA framework, the full raw dataset remains the preferred choice.

5.3 LDA decision-boundary plots

To visualize the geometry of the LDA rule, pairwise decision-boundary plots were constructed using the dataset `milk_no_out_yj`. This dataset is not the best-performing one in terms of LOOCV accuracy, but it provides a cleaner view of the separation pattern after reducing the influence of the most extreme observations and stabilizing marginal behavior through transformation. Since LDA assumes a common covariance matrix across groups, the resulting decision boundaries are linear in each two-variable projection.

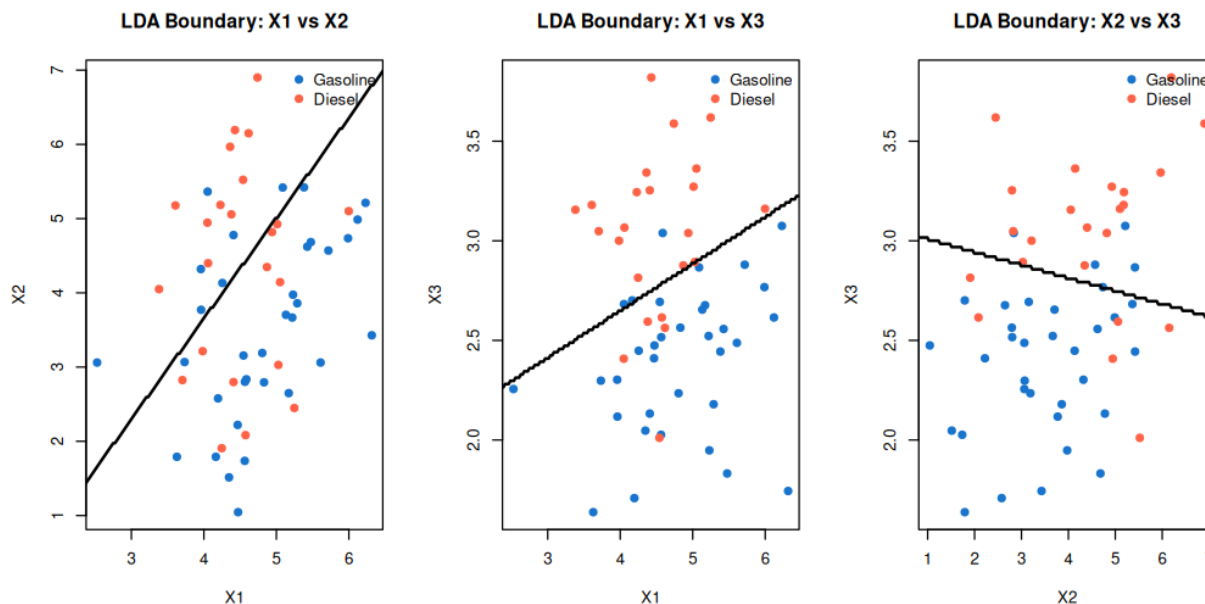


Figure 5.1: Pairwise LDA decision boundaries based on `milk_no_out_yj`.

Figure 5.1 shows that the fitted boundaries are straight lines in all three pairwise projections, as expected under LDA. The X_1 – X_2 boundary has a relatively steep positive slope, while the X_1 – X_3 boundary also slopes upward but more moderately. In contrast, the X_2 – X_3 boundary has a mild negative slope. Across all three panels, considerable overlap remains between the gasoline and diesel observations, so the linear rule achieves only partial separation.

The visual pattern also suggests that separation is more evident in projections involving X_3 , whereas the groups are less cleanly separated in the X_1 – X_2 plane. Even so, the overlap visible in these plots is consistent with the LOOCV results in the previous section: LDA classifies gasoline reasonably well, but diesel remains harder to distinguish. These plots therefore provide a useful geometric interpretation of why LDA yields only moderate classification performance for this dataset.

5.4 QDA: cross-validated classification performance

We next compare quadratic discriminant analysis (QDA) across the same four dataset versions using leave-one-out cross-validation. Unlike LDA, QDA allows the two fuel-type groups to have different covariance matrices, and hence produces a quadratic classification rule. As before, performance is summarized through the LOOCV confusion matrix, overall accuracy, and class-wise misclassification rates.

Table 5.2: LOOCV performance of QDA under different preprocessing choices

Dataset	GG	GD	DG	DD	Accuracy	Gasoline error	Diesel error
<code>milk</code>	33	3	5	18	0.8644	0.0833	0.2174
<code>milk_yj</code>	33	3	6	17	0.8475	0.0833	0.2609
<code>milk_no_out</code>	31	3	8	15	0.8070	0.0882	0.3478
<code>milk_no_out_yj</code>	31	3	6	17	0.8421	0.0882	0.2609

GG = Gasoline classified as Gasoline, GD = Gasoline classified as Diesel, DG = Diesel classified as Gasoline, DD = Diesel classified as Diesel.

Among all four versions, the full raw dataset `milk` gives the best QDA performance, with the highest LOOCV accuracy and the lowest diesel error rate. The transformed dataset `milk_yj` performs slightly worse, though it remains competitive. Removing the two influential gasoline observations causes a much larger drop in accuracy, especially through a deterioration in diesel classification. Applying Yeo–Johnson transformation after outlier removal improves QDA relative to `milk_no_out`, but still does not recover the performance of the raw-data fit.

Overall, the QDA results follow the same general pattern as LDA, but at a higher level of accuracy. In particular, QDA classifies diesel more effectively than LDA across all comparable settings. Thus, within the QDA framework as well, the full raw dataset remains the preferred choice for classification.

5.5 QDA decision-boundary plots

To visualize the geometry of the QDA rule, pairwise decision-boundary plots were again constructed using the dataset `milk_no_out_yj`. As in the LDA case, this dataset is used mainly for clearer visualization after reducing the influence of the most extreme observations and applying transformation. Since QDA allows the two groups to have different covariance matrices, the resulting decision boundaries are no longer restricted to be linear.

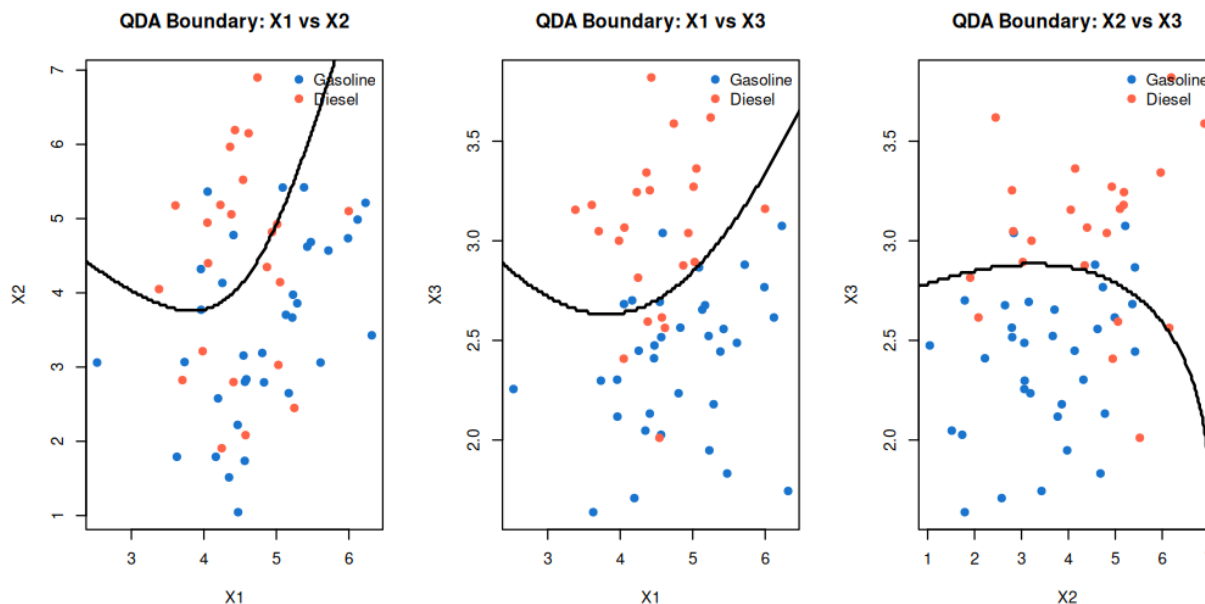


Figure 5.2: Pairwise QDA decision boundaries based on `milk_no_out_yj`.

Figure 5.2 shows that the fitted boundaries are curved in each pairwise projection, reflecting the greater flexibility of QDA. In the X_1 – X_2 plane the boundary bends noticeably, while in the X_1 – X_3 and X_2 – X_3 planes the separation pattern is also distinctly non-linear. These plots suggest that the covariance structure differs enough across the two groups for a quadratic rule to adapt more closely to the observed data cloud than a linear rule.

As in the LDA plots, substantial overlap remains between gasoline and diesel observations, so perfect separation is not possible. Nevertheless, the curved boundaries capture the group structure more flexibly, especially in projections involving X_3 . This is consistent with the LOOCV results of the previous section, where QDA achieved lower diesel misclassification and better overall accuracy than LDA. Thus, the decision-boundary plots provide a geometric explanation for the stronger predictive performance of QDA in this dataset.

Chapter 6

PCA and Factor Analysis

6.1 Suitability for Principal Component and Factor Analysis

Before applying principal component analysis and factor analysis, it is useful to check whether the three cost variables exhibit enough common correlation structure. For this purpose, we report Bartlett’s test of sphericity and the Kaiser–Meyer–Olkin (KMO) measure for both the raw data and the Yeo–Johnson transformed data.

Table 6.1: Preliminary checks for PCA and FA suitability

Dataset	KMO (overall MSA)	Bartlett’s test
Raw data (<code>milk</code>)	0.41	$\chi^2 = 17.106$, $p = 0.0007$
Yeo–Johnson data (<code>milk.yj</code>)	0.44	$\chi^2 = 12.315$, $p = 0.0064$

Bartlett’s test rejects the null hypothesis of an identity correlation matrix for both datasets, so the variables are not mutually uncorrelated. However, the KMO values are low in both cases. This suggests that although some shared structure is present, the data are only weakly suited for factor analysis. Therefore, PCA is more defensible as the primary dimension-reduction tool, while FA should be treated as exploratory only.

6.2 Principal Component Analysis

Principal component analysis was applied to the standardized variables X_1, X_2, X_3 . The first two components explain about 84.2% of the total variation, so a two-dimensional summary is adequate for descriptive purposes.

Table 6.2: Variance explained by the principal components

Component	Standard deviation	Proportion of variance	Cumulative proportion
PC1	1.2091	0.4873	0.4873
PC2	1.0315	0.3547	0.8420
PC3	0.6886	0.1580	1.0000

Table 6.3: Loadings of the first two principal components

Variable	PC1	PC2
X_1	-0.4564	0.7367
X_2	-0.7296	0.0111
X_3	-0.5093	-0.6762

PC1 is mainly a common cost dimension, with all three variables contributing substantially. PC2 is driven primarily by a contrast between X_1 and X_3 , while X_2 contributes very little to this direction. Since the signs of principal components are arbitrary, interpretation should be based on relative magnitude rather than sign.

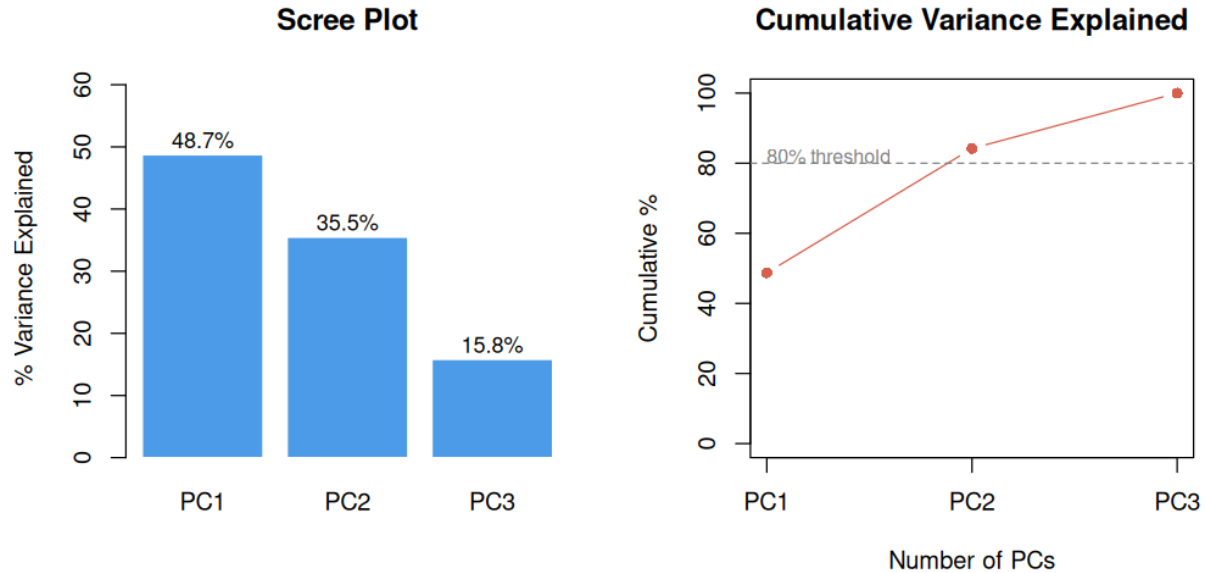


Figure 6.1: Scree plot and cumulative proportion of variance explained.

The scree plot also supports retaining two components: the third component contributes relatively little additional information, whereas the first two already capture most of the total variability.

6.3 Exploratory Factor Analysis

Since the preliminary diagnostics suggested only weak suitability for factor analysis, FA is used here in a limited exploratory sense. Parallel analysis suggested retaining two factors, and a two-factor maximum likelihood solution was fitted. For interpretation, the varimax rotation is reported because it gives the clearest loading pattern.

Table 6.4: Varimax-rotated two-factor solution

Variable	Factor 1	Factor 2	Communality	Uniqueness
X_1		0.690	0.4778	0.5222
X_2	0.562	0.510	0.5752	0.4248
X_3	0.700		0.4940	0.5060

Table 6.5: Variance explained by the rotated factors

Factor	SS loadings	Proportion of variance	Cumulative proportion
Factor 1	0.8069	0.2690	0.2690
Factor 2	0.7402	0.2467	0.5157

The two-factor solution explains about 51.6% of the total variation. Under varimax rotation, X_3 loads mainly on Factor 1, X_1 loads mainly on Factor 2, and X_2 shows moderate loadings on both factors. Thus, the factor structure is not especially clean, and the variables are only moderately represented by the fitted factors.

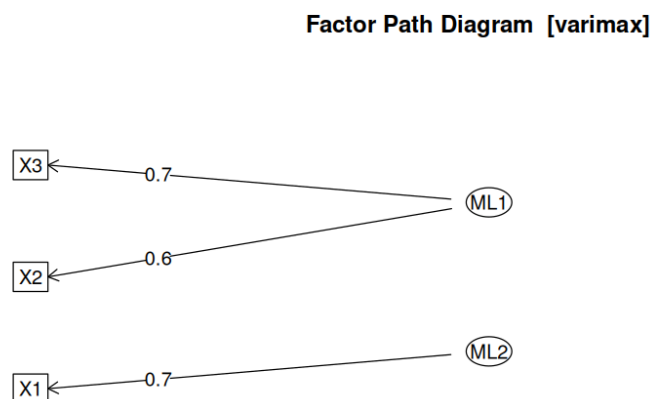


Figure 6.2: Factor path diagram for the varimax-rotated two-factor solution. Only the more prominent loadings are displayed.

Overall, the FA results are consistent with a weak two-factor description, but they do not provide a strong latent-variable interpretation. For this dataset, FA is best viewed as a secondary exploratory summary rather than the main dimension-reduction method.

Appendices

Appendix A

Normality Tests Used

Let $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ denote a sample, with sample mean $\bar{\mathbf{x}} = n^{-1} \sum_{i=1}^n \mathbf{x}_i$ and sample covariance matrix

$$S = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top.$$

For univariate tests, write the observations as $x_1, \dots, x_n$, with order statistics $x_{(1)} \leq \dots \leq x_{(n)}$. The null hypothesis is always

$$H_0 : \text{the sample arises from a normal distribution}$$

in the univariate case, or from a p -variate normal distribution in the multivariate case.

A.1 Univariate Normality Tests

A.1.1 Shapiro–Wilk Test

Let $\mathbf{z}_{(1)}, \dots, \mathbf{z}_{(n)}$ be the order statistics of an i.i.d. $N(0, 1)$ sample, and write

$$\mathbf{m} = E(\mathbf{z}_{(\cdot)}), \quad V = \text{Cov}(\mathbf{z}_{(\cdot)}).$$

The Shapiro–Wilk statistic is

$$W = \frac{(\mathbf{a}^\top \mathbf{x}_{(\cdot)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \mathbf{a} = \frac{V^{-1} \mathbf{m}}{\sqrt{\mathbf{m}^\top V^{-1} V^{-1} \mathbf{m}}}.$$

Thus W compares the sample order-statistic pattern to the one expected under normality. Values of W close to 1 support H_0 , while small values indicate non-normality.

A.1.2 Anderson–Darling Test

Let

$$\hat{z}_i = \frac{x_i - \hat{\mu}}{\hat{\sigma}}, \quad \hat{F}_0(x) = \Phi\left(\frac{x - \hat{\mu}}{\hat{\sigma}}\right),$$

where $\hat{\mu} = \bar{x}$ and $\hat{\sigma}$ is the sample standard deviation. The Anderson–Darling statistic is

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n (2i-1) \left[\log \Phi(\hat{z}_{(i)}) + \log(1 - \Phi(\hat{z}_{(n+1-i)})) \right].$$

This is a weighted empirical-distribution test, with extra emphasis on the tails. Large values of A^2 lead to rejection of H_0 .

A.1.3 Lilliefors Test

The Lilliefors statistic is the Kolmogorov-type distance

$$D_L = \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - \hat{F}_0(x)|,$$

where $\hat{F}_n$ is the empirical cdf and

$$\hat{F}_0(x) = \Phi\left(\frac{x - \hat{\mu}}{\hat{\sigma}}\right)$$

uses parameters estimated from the same sample. Thus it is a goodness-of-fit test to normality with unknown mean and variance. Large values of D_L indicate non-normality.

A.2 Multivariate Normality Tests

A.2.1 Mardia's Skewness and Kurtosis

Define the squared Mahalanobis distances

$$d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top S^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}), \quad i = 1, \dots, n.$$

Mardia's multivariate skewness and kurtosis are

$$b_{1,p} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \left[(\mathbf{x}_i - \bar{\mathbf{x}})^\top S^{-1}(\mathbf{x}_j - \bar{\mathbf{x}}) \right]^3,$$

and

$$b_{2,p} = \frac{1}{n} \sum_{i=1}^n d_i^4.$$

Under H_0 ,

$$\frac{n b_{1,p}}{6} \stackrel{a}{\sim} \chi_{\frac{p(p+1)(p+2)}{6}}^2, \quad \frac{b_{2,p} - p(p+2)}{\sqrt{8p(p+2)/n}} \stackrel{a}{\sim} N(0, 1).$$

Hence large skewness or kurtosis departures indicate violation of multivariate normality.

A.2.2 Henze–Zirkler Test

Let

$$\mathbf{y}_i = S^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}}), \quad i = 1, \dots, n,$$

so that the sample is affinely standardized. The Henze–Zirkler statistic is based on a weighted L^2 distance between the empirical characteristic function

$$\phi_n(\mathbf{t}) = \frac{1}{n} \sum_{j=1}^n e^{i\mathbf{t}^\top \mathbf{y}_j}$$

and the characteristic function of $N_p(\mathbf{0}, I_p)$,

$$\phi_0(\mathbf{t}) = e^{-\frac{1}{2}\mathbf{t}^\top \mathbf{t}}.$$

Formally,

$$HZ_{n,\beta} = n \int_{\mathbb{R}^p} |\phi_n(\mathbf{t}) - \phi_0(\mathbf{t})|^2 w_\beta(\mathbf{t}) d\mathbf{t},$$

where w_β is a Gaussian weight. Thus the test measures global discrepancy from multivariate normality. Large values reject H_0 .

A.2.3 Royston Test

Royston’s test is an omnibus multivariate extension of the Shapiro–Wilk/Shapiro–Francia family. For each margin $j = 1, \dots, p$, let W_j denote the corresponding univariate normality statistic, and transform it to an approximately standard normal score

$$Z_j = \Gamma_n(W_j).$$

The vector $\mathbf{Z} = (Z_1, \dots, Z_p)^\top$ is then combined using a dependence adjustment,

$$R = \mathbf{Z}^\top \hat{\Omega}^{-1} \mathbf{Z},$$

where $\hat{\Omega}$ estimates the correlation structure among the transformed marginal statistics. Under H_0 , R is approximately chi-square. Large values indicate departure from multivariate normality.

A.2.4 Doornik–Hansen Test

The Doornik–Hansen procedure first affinely standardizes and approximately orthogonalizes the sample. For each transformed component, univariate skewness and kurtosis are computed and then normalized to yield approximately standard normal variables, say

$$Z_{1j} \quad \text{and} \quad Z_{2j}, \quad j = 1, \dots, p.$$

The omnibus statistic is

$$DH = \sum_{j=1}^p (Z_{1j}^2 + Z_{2j}^2),$$

which is approximately distributed as χ_{2p}^2 under H_0 . Thus the test jointly aggregates skewness- and kurtosis-type departures across all transformed components.

A.3 Interpretation

The univariate tests assess marginal normality of each variable separately, whereas the multivariate tests assess the joint elliptical-Gaussian structure of the full vector. Since later inferential procedures rely primarily on *groupwise* multivariate normality, these tests will be applied separately to the gasoline and diesel groups, with pooled-data results used only as secondary diagnostics.